

Servopneumatic drive system for robot welding guns

FESTO

Manual

Servopneumatic drive
system for robot welding
guns

for controller type
CPX-SB-LT-WELD-SA

Variant

- CPX servobox
- 7 axis concept
- Fieldbus ProfiNet FOC

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1 Introduction

1.1 Important note on this documentation

This documentation contains general basic information on mounting, installation and commissioning of a controlled welding tong drive.



This documentation contains the description of the subsequent variants.

The documentation is only to be used by instructed and experienced persons. It does not contain the detail required of complete technical documentation for a regular mass-produced product.



Information on the CPX node used can be found in the corresponding description, e.g. P.BE-CPX-PNIO-DE for the CPX-M-FB35 with network protocol PROFINET IO.

PROFINET IO® and Windows® are registered trademarks of the respective trademark owners in certain countries.

1.2 Safety

This documentation contains instructions on the possible dangers which can occur if the product is incorrectly used.



Warning

Electric shock (protection against direct and indirect contact).

- Use only PELV circuits in accordance with IEC/DIN EN 60204-1 (protective extra-low voltage, PELV) for the electrical power supply.
- Observe also the general requirements for PELV circuits in accordance with IEC/DIN EN 60204-1.
- Use only power sources which guarantee reliable electrical isolation of the operating voltage in accordance with IEC/DIN EN 60204-1.



Warning

Accidental movements of the connected actuator technology and uncontrolled movements of loose tubing can cause injury to people or material damage.

Before carrying out mounting, installation and maintenance work, switch off the following:

- the compressed air supply
- operating and load voltage supplies.

Use all system components only as intended and in their original status.

The drives move dynamically with great force. Make sure that nobody can enter the sphere of influence of the drives while the system is connected to energy sources.

Errors in installation or parameterisation can cause injury to people and damage to property. Operate the system only if it has been professionally installed and parameterised.

Attention! Incorrect control of the welding guns with the PC controller can cause personal injury or material damage

1.3 The servopneumatic drive system from Festo - an overview

Pneumatic drives for welding gun for resistance point welding are used worldwide. Advantages of the pneumatics are sturdiness, simple commissioning, small installation space as well as low acquisition and maintenance costs.

With servopneumatic drive solutions, controlled positioning procedures and force profiles are also possible – the existing advantages of the pneumatics remain.

Major features of the servopneumatic solution are:

- Free and fast positioning of the electrodes.
This permits a reduction of the cycle time and avoidance of obstacles.
- Gentle fitting of the electrodes.
This increases the quality of the welding point and the useful life of the electrodes. Noise reduction through gentle fitting is also a relevant point.
- Fast and controlled force build-up.
This permits high reproducibility of force, reduction of the cycle time and consistent quality of the welding points.
- Freely adjustable tong balance for weight equalization of the welding gun.
As a result, the mechanical load on the metal sheets is minimised and can be adapted to the space situation of the robot.
- Lower energy consumption compared to classical solutions.

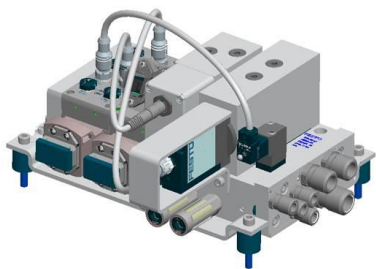
The servopneumatic system is a complete package made up of various individual components. The main cylinders with various strokes and diameters allow for different force areas and can be used for various installation conditions in C- and X-gun. Compensating cylinders are also offered in different dimensions. The valve control block is used to control the two cylinders. A controller with the firmware permits force and position control of the main cylinder, diagnostics and all functions necessary for commissioning, operation and communication with the higher-order controller. Through the fieldbus module, data can be exchanged with the higher-order system, e.g. the robot control. All fieldbus systems established in the automobile industry are thereby supported (e.g. Interbus CU and FOC, Profibus, ProfiNet CU and FOC, DeviceNet, ...). The supplied software is used for project administration, commissioning, parameterisation and diagnostics. Various control algorithms and protocols for welding gun drives are implemented: from point-to-point controls to innovative track-controlled 7-axis concepts. Accessory parts, such as hoses, cables or filters, round out the overall package.

The system from Festo is already being used worldwide at leading customers. Festo supports its customers through worldwide service. Therefore, besides the products, Festo offers training and service packages.

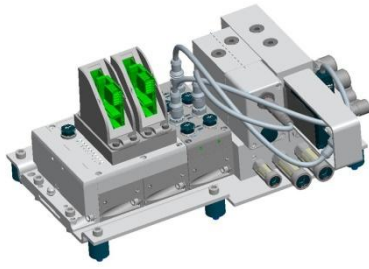
2 Components

2.1 Controllers

1



2

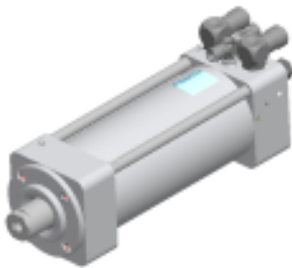


No.	Component	Designation	Part No.
1	Controller with valve block (ProfiNet FOC)	CPX-SB-LT-WELD-SA	1393386
2	Controller with valve block (Interbus FOC)	CPX-SB-LT-WELD-FB21-SA	2310097

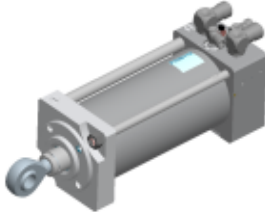
Table 2.1: Controller

2.2 Main cylinder

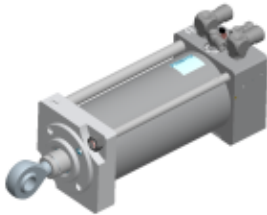
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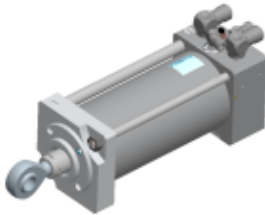
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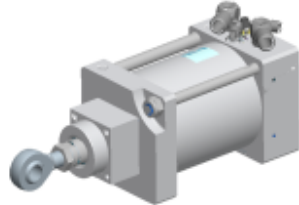
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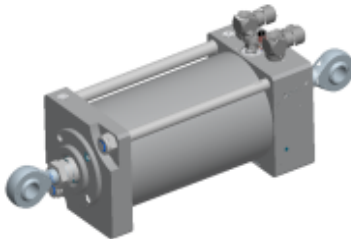
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5



6



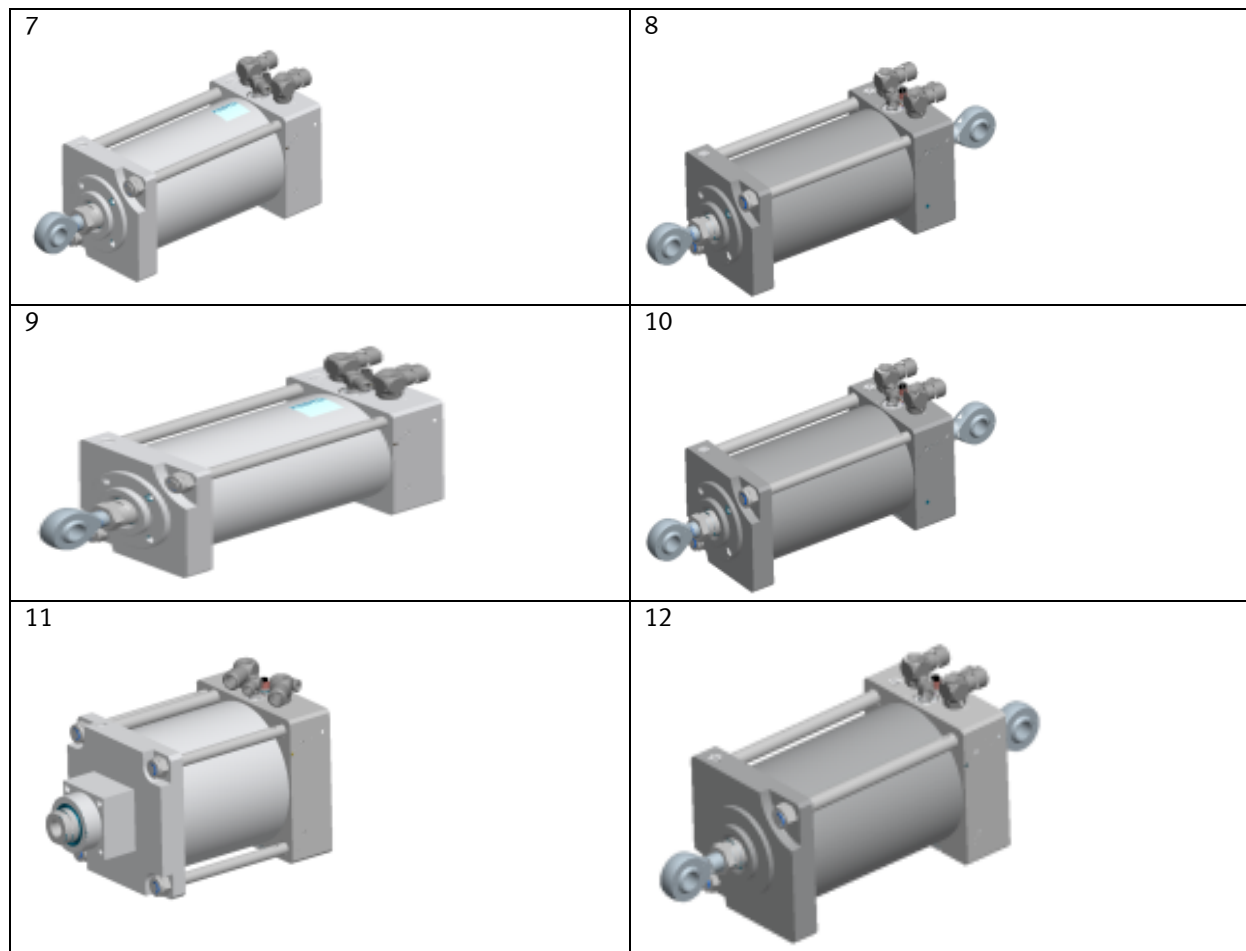


Figure 1: Main cylinder

No.	Component	Designation	Part No.
1	Main cylinder dia. 90 mm, stroke 150 mm	D-90-SPA-150-SA/C	1461797
2	Main cylinder dia. 100 mm, stroke 160 mm	D-100-SPA-160-SA/C	1385359
3	Main cylinder dia. 100 mm, stroke 184 mm	D-100-SPA-184-SA/C	2196582
4	Main cylinder dia. 100 mm, stroke 248 mm	D-100-SPA-248-SA/C	3068894
5	Main cylinder dia. 125 mm, stroke 90 mm	D-125-SPA-90-SA/X	1410253
6	Main cylinder dia. 125 mm, stroke 160 mm	D-125-SPA-160-SA/X	1447790
7	Main cylinder dia. 125 mm, stroke 160 mm	D-125-SPA-160-SA/C	1439658
8	Main cylinder dia. 125 mm, stroke 184 mm	D-125-SPA-184-SA/C	2195510
9	Main cylinder dia. 125 mm, stroke 248 mm	D-125-SPA-248-SA/C	1933103
10	Main cylinder dia. 140 mm, stroke 150 mm	D-140-SPA-150-SA/X	1477510
11	Main cylinder dia. 160 mm, stroke 90 mm	D-160-SPA-90-SA/X	1450035
12	Main cylinder dia. 160 mm, stroke 150 mm	D-160-SPA-150-SA/X	2277634

Table 2.2: Main cylinder

2.3 Compensating cylinder

The following table shows the common compensating cylinders. Additional compensating cylinders can be made available on request.

No.	Component	Designation	Part No.
1	Compensating cylinder Compensating C-gun	ADN-50-40-A-P-A	536316
2	Compensating cylinder Compensating X-gun	D-63-20-P-SA	526604

Table 2.3: Compensating cylinder

2.4 Connecting cables

No.	Component	Designation	Part No.
1	Connecting cable, drive connection to the main cylinder 1A1 Straight plug - angled socket	NEBU-M12G8-K-1,1-M12W8-SA	566056
2	Connecting cable, drive connection to the main cylinder 1A1 Straight plug - straight socket	NEBU-M12G8-K-1,1-M12G8	572919
3	Connecting cable, comp. connection to the MPYD pressure regulating valve 2V1	NEBU-M9G8-K0,35-M12G5	573908
4	Connecting cable, valve connection to the MPYE servo valve 1V1	NEBU-M12G4-K-0,35-M12W4-SA	573909
5	Connecting cable, connection output 0 to the switching valve 1V2	KMEB-1-24-M12G2-0,45-LED	573910

Table 2.4: Connecting cables

2.5 Accessories and spare parts

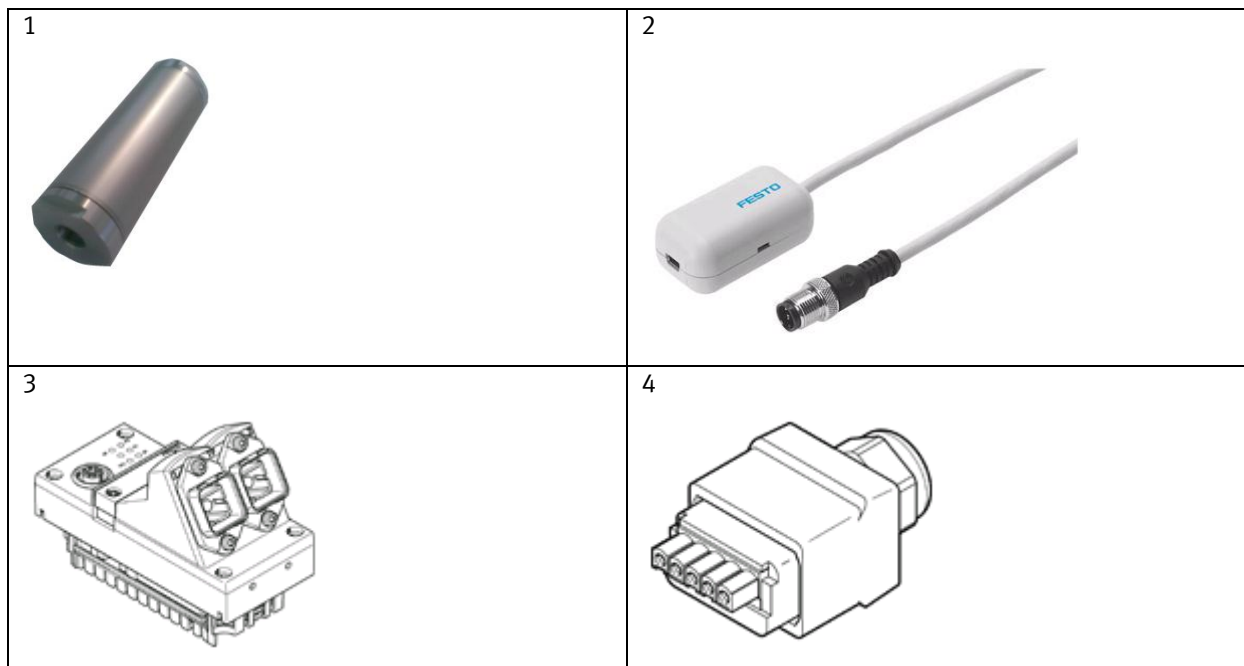


Figure 2: Accessories

No.	Component	Designation	Part No.
1	Filter (5 µm)	Inline filter LF-3/8-C-SA	1563346
2	USB serial adapter M12	NEFC-M12G5-0.3-U1G5	547432
3	ProfiNet bus node FOC	CPX-M-FB35	548749
4	Aida push-pull plug connector	NECU-M-PPG5-C1	563059

Table 2.5: Accessories and spare parts

3 Controllers

3.1 Pin allocation

The illustration below shows the connecting elements of the controller.

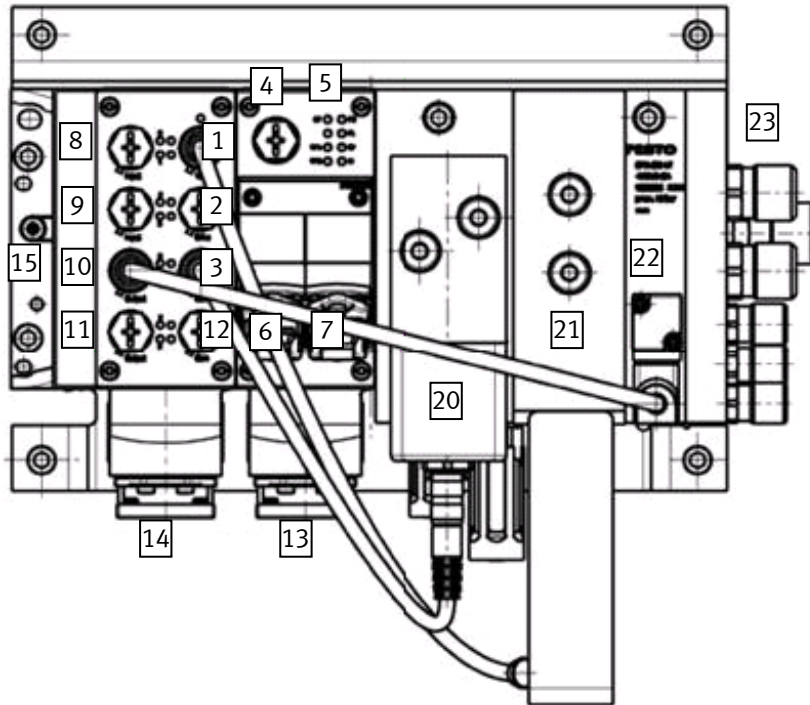


Figure 3: Controller CPX-SB-LT-WELD-SA

Connection	Designation	Comment
1	Valve	Connection to the proportional directional control valve MPYE 1V1
2	Drive	Connection to the main cylinder 1A1
3	Equalizer	Connection to the pressure regulating valve MPYD 2V1
4	Com	RS232 interface to PC
5	Display	Fieldbus status display
6	Fieldbus	ProfiNet FOC
7	Fieldbus	ProfiNet FOC
8	Input 0+1	2 digital inputs 24 V
9	Input 2+3	2 digital inputs 24 V
10	Output 0+1	2 digital outputs 24 V
11	Output 2+3	2 digital outputs 24 V
12	Ext sensor	2 analogue inputs
13	Supply 1	AIDA power supply 1
14	Supply 2	AIDA power supply 2
15	PE	Connection for protective conductor

Table 3.1: Electrical connections



: The controller must be earthed with the intended PE connection.
All electrical connections must be sealed with appropriate protective caps.

Valve block

Connection	Designation	Comment
20	2V1 (MPYD)	Proportional pressure regulator valve
21	1V1 (MPYE)	Proportional directional control valve
22	1V2 (MSEB)	Control of shut-off valves 1V3 and 1V4 in the main cylinder
23	Ports	Pneumatic connections

Connection 1 proportional directional control valve 1V1 MPYE (valve)

Pin	Signal	Pin	Signal
1	24 V supply	3	Valve setpoint specification 0 ... 10 V
2	0 V supply	4	GND

Connection 2 main cylinder 1A1 (Drive)

Pin	Signal	Pin	Signal
1	12 V supply to pressure sensors	5	Position signal
2	–	6	Pneumatic signal V
3	10 V reference voltage	7	Pneumatic signal R
4	GND position signal	8	GND pressure sensors

Connection 3 equalization valve 2V1 MPYD (comp.)

Pin	Signal	Pin	Signal
1	24 V supply	4	GND
2	0 V supply	5	Actual value equalization
3	Setpoint value equalization		

Connection 8 + 9 digital inputs (input)

Pin	Signal	Pin	Signal
1	24 V	3	0 V
2	Input X+1	4	Input X

Connection 10+11 digital outputs (output)

Pin	Signal	Pin	Signal
1	–	3	0 V
2	Output X+1	4	Output X

Connection 12 ext sensor (COM)

Pin	Signal	Pin	Signal
1	24 V	4	Analogue input 0
2	Analogue input 1	5	PE
3	0 V		

Connection 13+14 power and load voltage supply

Pin	Signal	Pin	Signal
1	US1+	4	US2-
2	US1-	5	PE
3	US2+		

3.2 Fieldbus node

The servopneumatic welding gun system supports various fieldbus protocols. The base of this design is the CPX fieldbus node FB35 – Profinet FOC (fibre-optic cable) or FB21 Interbus FOC.

For a detailed description of the function of the fieldbus node, we refer you to the manual CPX-M-FB35 or CPX-M-FB21.

3.2.1 LED indicators FB34/FB35

LED indicators	
Bus-specific	NF = Network fault
	TP1 = Network active port 1
	TP2 = Network active port 2
Product-specific	M = Modify, parameterisation
	PL = Load supply
	PS = Electronics supply, sensor supply
	SF = System fault

Table 3.2: LED indicators

3.2.2 LED indicators FB20/FB21

LED indicators	
INTERBUS-specific	BA = Bus active
	FO1 = Fibre-optic cable 1
	FO2 = Fibre-optic cable 2
	RC = Remote bus check
	RD = Remote bus disable
CPX-specific	M = Parameterisation
	SF = System fault
	US1 = Electronics supply, sensor supply
	US2 = Load voltage supply

Table 3.3: LED indicators

4 Mounting

4.1 General instructions on mounting and dismantling



Warning

Accidental movements of the connected actuators and uncontrolled movements of loose tubing lines can cause injury to people or material damage.

Before carrying out mounting, installation and maintenance work, switch off the following:

- the compressed air supply
- operating and load voltage supplies.

4.2 Installation dimensions of the controller

The controller can be fastened to a flat surface with the mounting plate.

The following drawing shows the mounting surface and dimensions for the 4 threaded holes of screw size M6. The controller must be fastened to the welding gun with the 4 rubber feet provided.

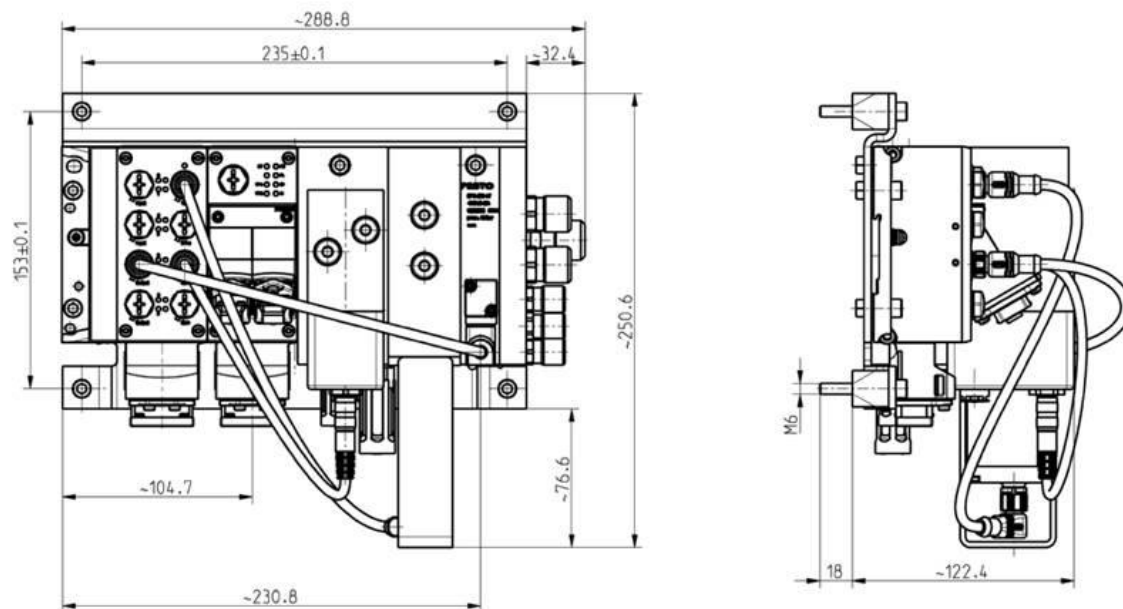


Figure 4: Dimensional drawing CPX-SB-LT-WELD-SA

4.3 Controller replacement

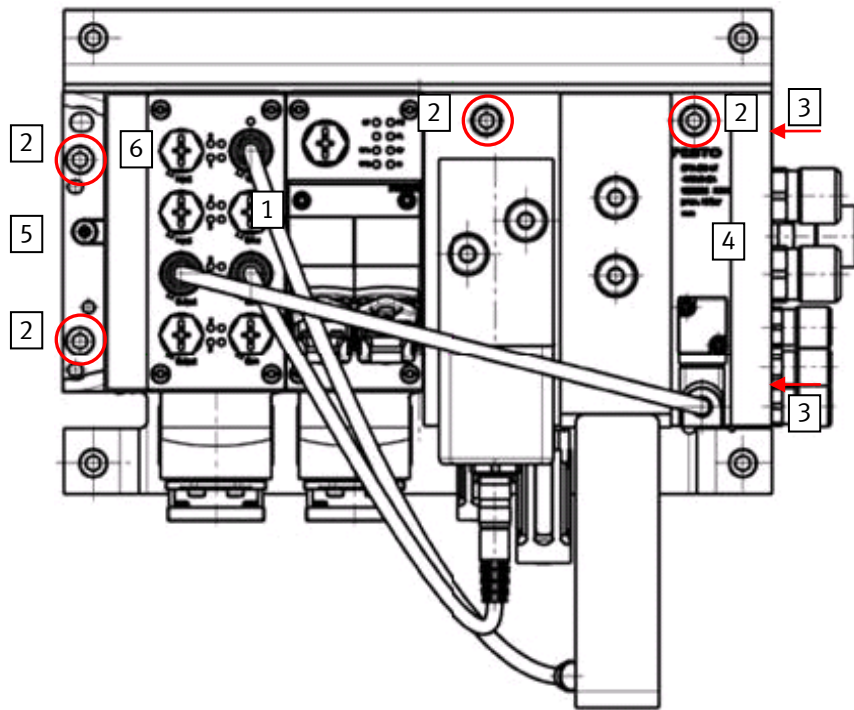


Figure 5: Controller CPX-SB-LT-WELD-SA

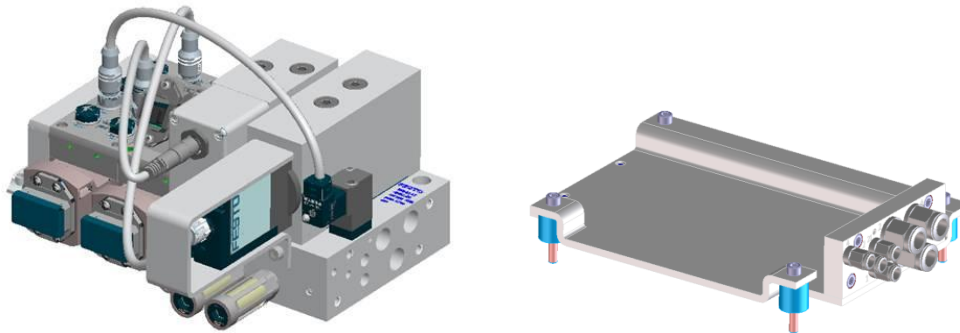


Figure 6: Controller and mounting plate

It is possible to replace the controller with the valves without disconnecting the tubing package. A mounting plate with fast-change fixture is used.

Screw(s)	Tightening torque
2	8 Nm
3	8 Nm
5	1 Nm

Table 4.1: Tightening torques of screws

Action	Action
1	Loosen cable to the main cylinder [1]
2	Loosen cable to the thermal contact of the transformer [6]
3	Disconnect PE-connection [5]
4	Loosen 4 x lateral screws of the sub-base [3]
5	Loosen 4 x controller screws [2]
6	Replace controller. Note: During mounting, contamination must not enter the system!
7	Place seal between mounting plate and controller [4]
8	Tighten 4 x lateral screws of the sub-base [3]
9	Tighten 4 x controller screws [2]
10	Connect cable to the main cylinder [1]
11	Connect cable to the thermal contact of the transformer [6]
12	Connect PE connection [5]

Table 4.2: Approach to controller replacement

5 Installation

5.1 Pneumatic circuit diagram

The illustration shows the pneumatic circuit diagram. A pneumatic circuit is integrated in the main cylinder 1A1. The function of the additional switch is both to shut off the main connections through the valves 1V2, 1V3 and 1V4 at connection 21 and perform exhaust via the valve 1V5 at connection 1 of the main cylinder, if the supply pressure is switched off.

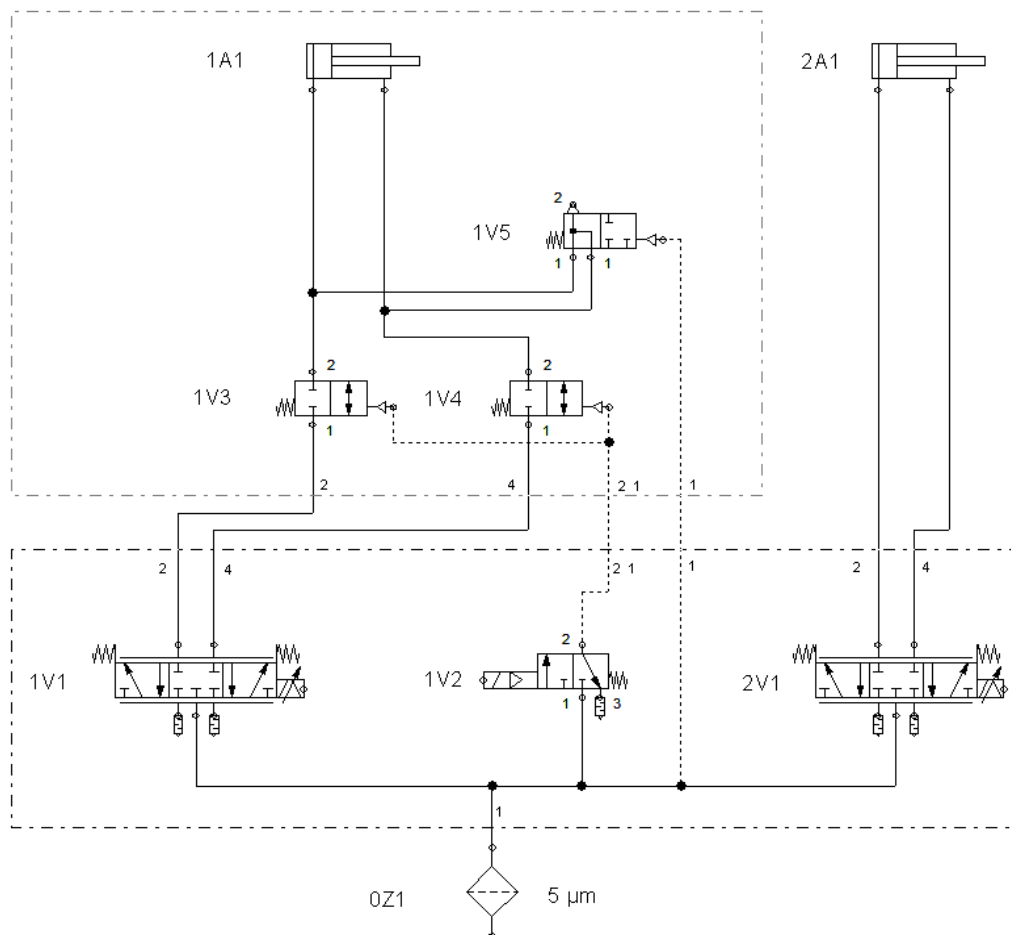


Figure 7: Pneumatic circuit diagram

Designation	Description
1V1	Proportional directional control valve MPYE
1V2	Control of shut-off valves in the main cylinder MSEB
1V3	Shut-off valve integrated into the main cylinder
1V4	Shut-off valve integrated into the main cylinder
1V5	Exhaust valve integrated into the main cylinder
1A1	Main cylinder
2V1	Proportional pressure regulator valve MPYD
2A1	Compensating cylinder
OZ1	Filter (5 μ m)

Table 5.1: Allocation table

5.2 Pneumatic installation

5.2.1 Connection of operating pressure and inline filter

A 5 µm inline filter must be installed between the connection of the main air supply to the welding gun and connection 1 of the main air supply to the mounting plate.

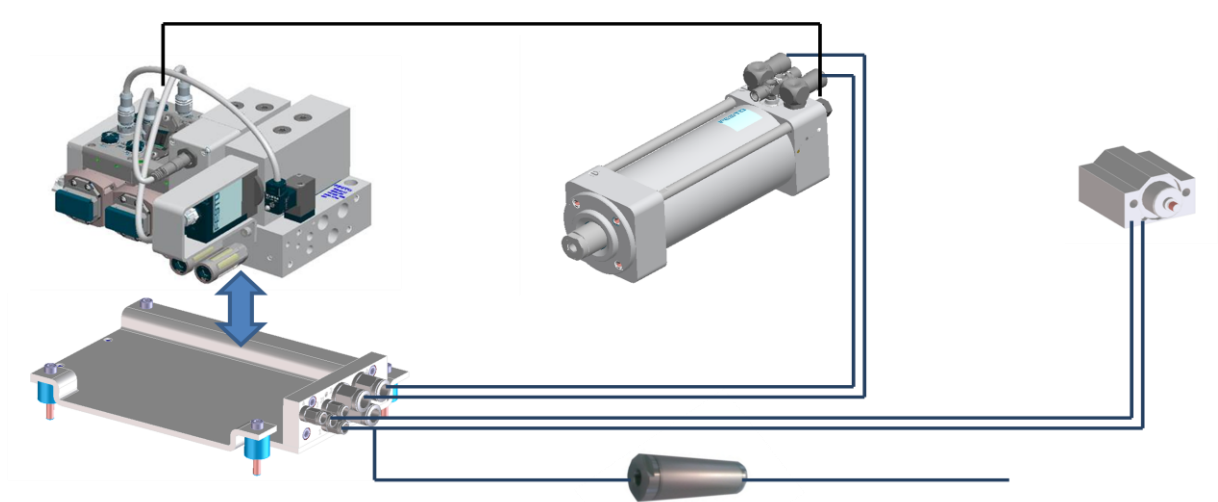


Figure 8: Overview of pneumatic connection

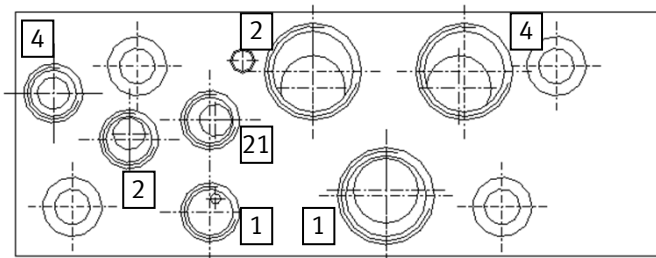


Figure 9: Pneumatic connections of the mounting plate

Connection	Fitting	Description
1	G3/8	Main air supply
1	G1/8	Exhaust valve 1V5 (main cylinder connection 1)
21	G1/8	Shut-off valve 1V3 /1V4 (main cylinder connection 21)
2	G3/8	Proportional directional control valve 1V1 (main cylinder connection 2)
4	G3/8	Proportional directional control valve 1V1 (main cylinder connection 4)
2	G1/8	Proportional-pressure regulator 2V1 (compensating cylinder 2 or 4)
4	G1/8	Proportional-pressure regulator 2V1 (compensating cylinder 4 or 2)

Table 5.2: Pneumatic connections of the mounting plate

5.2.2 Main cylinder

The tube couplings are numbered at the main cylinder and on the mounting plate. Make sure that the hoses are seated tight. The hoses to the main cylinder should be as short as possible.

Connection	Tubing inside diameter	Description
1	8 mm	Pilot pressure (exhaust main cylinder)
21	8 mm	Blocking function (blocking of main cylinder connection 2 + 4) US2 = 1 (operating pressure) US2 = 0 (pressure-free)
2	12/14 mm	Rear chamber (extend main cylinder)
4	12/14 mm	Front chamber (retract main cylinder)

Table 5.3: Main cylinder connections

5.3 Compensating cylinder

When connecting the compensating cylinder, note that a negative differential pressure is set for movement to the welding piece, and a positive differential pressure for movement to the stop. The active direction of the equalizer is determined through the tubing connection between the mounting plate and compensating cylinder and must be turned, dependent on the design X or C of the welding gun.

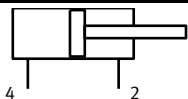
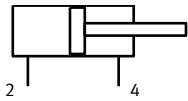
Design	Pneumatic connection
C-gun	
X-gun	

Table 5.4: Pneumatic connections, compensating cylinder

Connection	Tubing	Description, C-gun	Description, X-gun
2	8 mm	Front chamber	Rear chamber
4	8 mm	Rear chamber	Front chamber

Table 5.5: Compensating cylinder connections

6 Maintenance instructions

The main cylinders can be serviced by the operator. Due to the non-contacting displacement encoder, the main cylinder can be dismounted. For mounting, observe that tools must be used that permit easy mounting of the piston rod without damage to the seals and scrapers.

The lubrication intervals should not be more than 3 million welding points apart. To achieve high service lives, it is recommended that you protect the piston rod and seal from dust, sweat particles, etc., for example, by means of a protective sleeve.

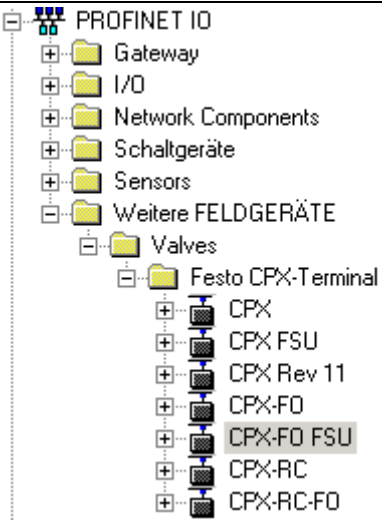
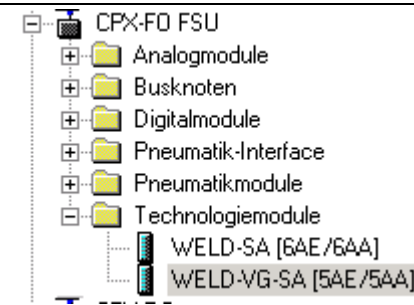
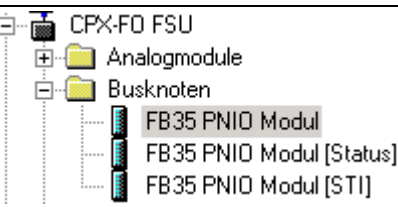
To achieve the service lives, Festo specifies lithium-saponified greases - NLGI class 2, e.g. CentoPlex 2EP/ type 301673, Centoplex 2EP (silicon-free grease)

Replacement of the 5 µm filter is recommended every 1 million welding points.

7 Integration of ProfiNet

7.1 Hardware configuration with Simatic S7

Here is shown how ProfiNet hardware configuration is displayed, using a Simatic S7 as an example. Configuration can also be made with other ProfiNet configuration tools. (e.g. Primary Setup Tool from Siemens)

No.	Action	Comment
1	HW Config open	Configure Ethernet ProfiNet IO system. 
2	ProfiNet IO CPX – insert FO FSU (FO = fibre optics) (FSU = Fast Startup)	
3	Insert at module space 1 with weld VG- SA [5AE/5AA]	
4	Insert module at module space 2 with FB35	

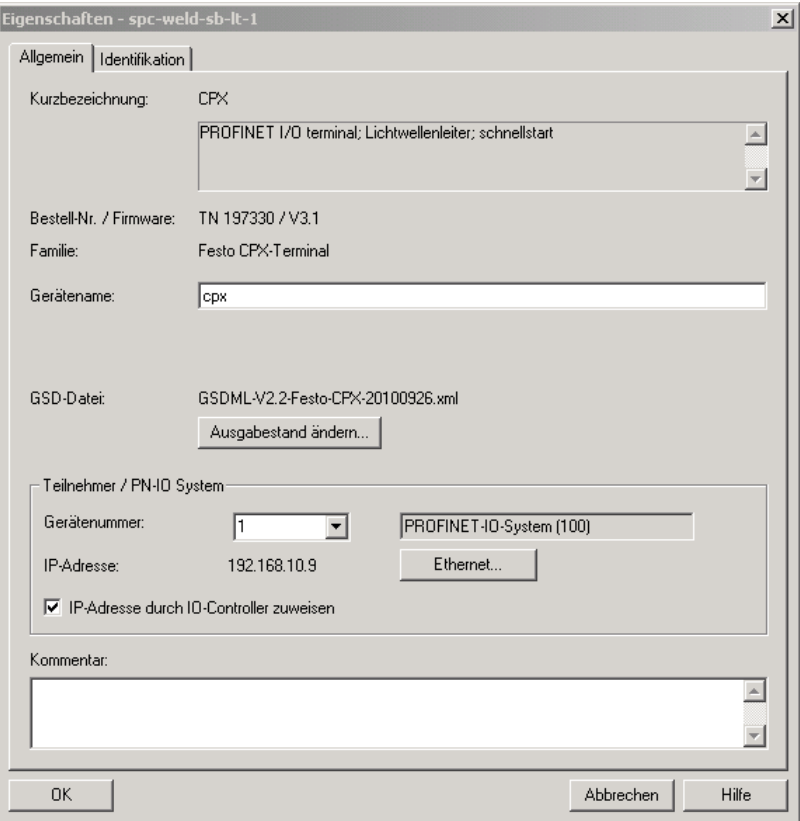
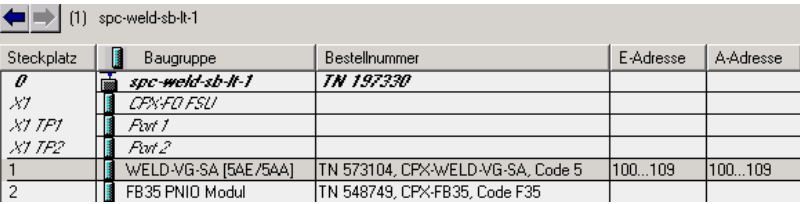
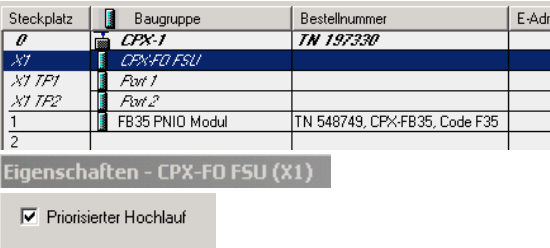
No.	Action	Comment
5	Assign device name Example “cpx” The IP address is customarily assigned by the robot	
6	View of CPX module “Welding gun controller”	
7	If the Fast Startup option is required, the “Prioritised running up” option must be set	

Table 7.1: Configuration example for ProfiNet

7.2 Edit IP addresses and ProfiNet name

With the WinSPZ tool, there is the possibility to set the ProfiNet project name and the IP address to the delivery status.

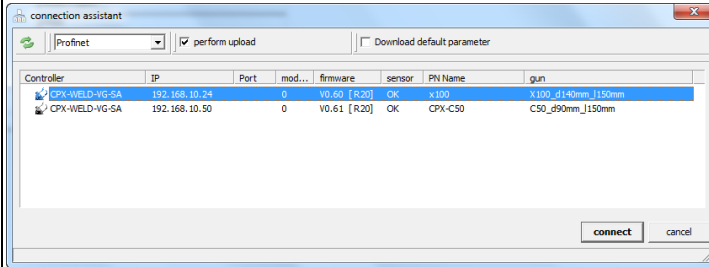
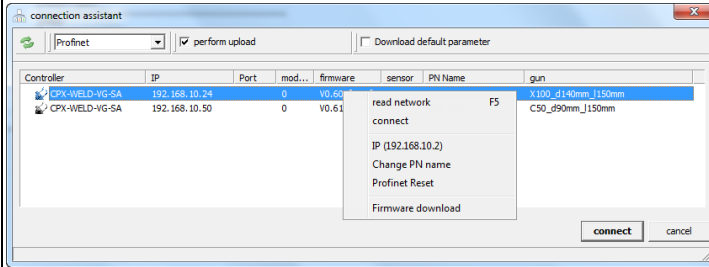
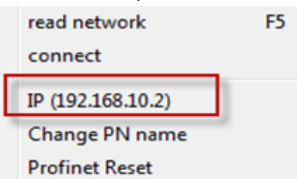
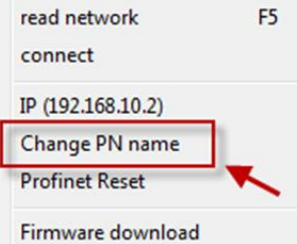
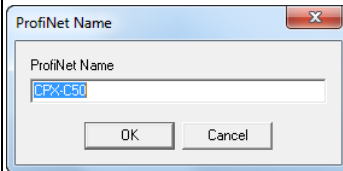
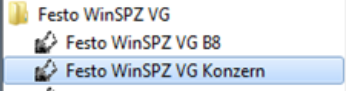
No.	Action	Comment
1	Connection	
2	Actuate right mouse button	 Note: Changes to the network configuration are only possible if no active ProfiNet connection exists between robots and servo-box.
3	IP address 192.168.10.2	
4	Change ProfiNet-Other project, name	With this menu option, the name on the ProfiNet can be changed.  Input ProfiNet name 
5		With this menu option, the ProfiNet controller is set to the factory settings. - ProfiNet project name deleted - IP address 0.0.0.0 

Table 7.2: Edit IP addresses and ProfiNet name

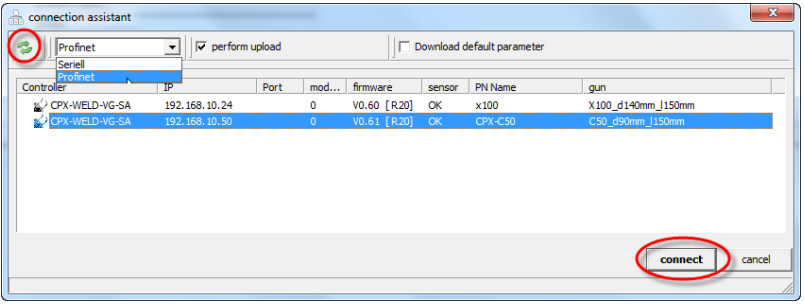
8 Commissioning

The steps for commissioning are explained in the following. The WinSPZ software is used for this purpose. The detailed explanation for operating the WinSPZ software is provided in the subsequent chapter

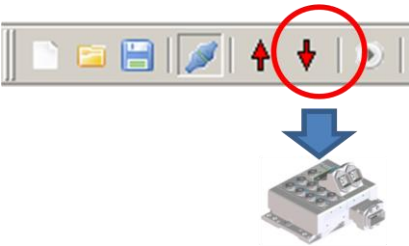
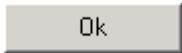
8.1 Overview of the commissioning steps

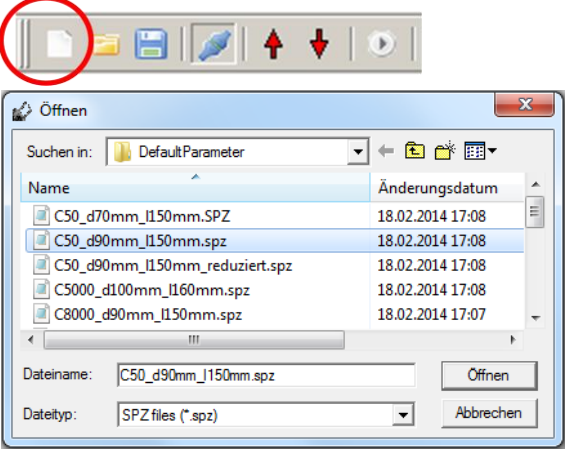
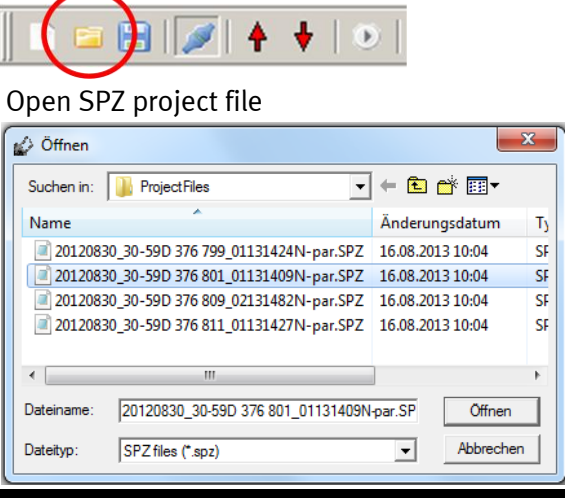
No.	Action	Comment
1	Start WinSPZ	Selection of the fieldbus protocol Selection of “VG Konzern” (Volkswagen Group) for the VW/AUDI MQB standard. 
2	Load base parameters	Load predefined parameter set for the respective welding gun type.
3	Set the welding gun parameters	Set basic data for the welding gun.
4	Perform axis calibration	Determine zero position, translation and maximum opening of the welding gun.
5	Perform force calibration	Scaling of the electrode force.
6	Set the welding gun parameters	Set or check additional parameters.
7	Set equalizer	Set and check the welding gun equalizer.
8	Check welding gun for proper function	Check positioning behaviour and force build-up of the welding gun.
9	Back up data	Save parameters to hard disk.

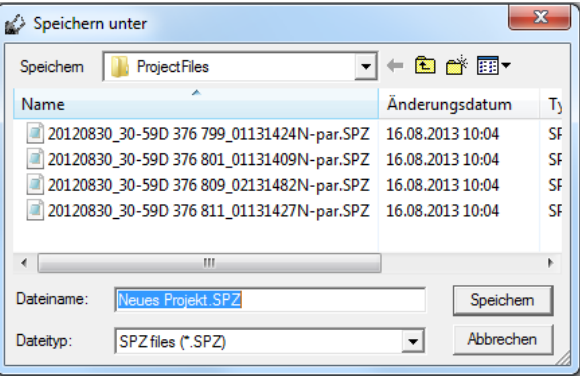
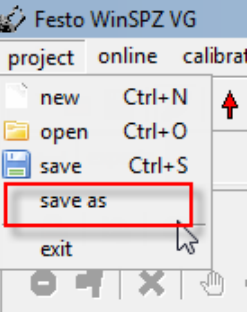
8.2 Establishing a connection to the controller

No.	Action	Comment
1	Connect cable to PC	CPX system serial: The serial interface is located on the bus node CPX system ProfiNet: Use Ethernet connection via system PC.
2	Open connection assistant	
3	Select serial or ProfiNet access type	The interface is searched for automatically. Actuate “Update”. 
3	Serial	The interface is searched for automatically and the found port entered.
4	ProfiNet	All welding gun systems found are displayed. Note: The subnet mask and the IP address range must agree with the network configuration.
5	Select controller and actuate “Connect”	 Note: If the ☐ “Perform upload” function is active, all parameters are read out of the controller.

8.3 General operating instructions

No.	Advice	Comment
1	Download Important! After each change of a parameter, “Download” must be actuated	Parameters are transmitted to the controller and written to the EEPROM 
2	Upload	Parameters are read out of the controller 
3	Axis calibration / force calibration	When the travel or force calibration is ended with OK, the parameters are transmitted automatically to the controller and saved 

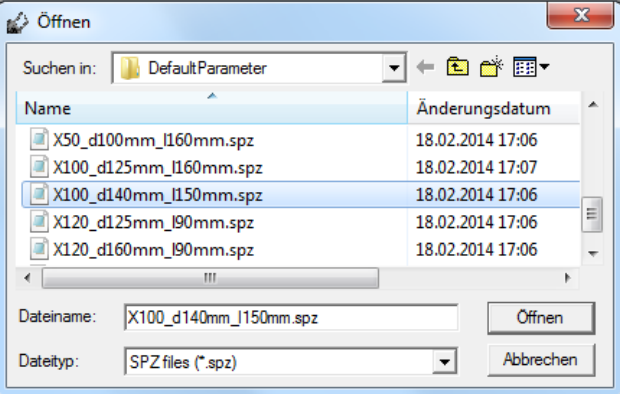
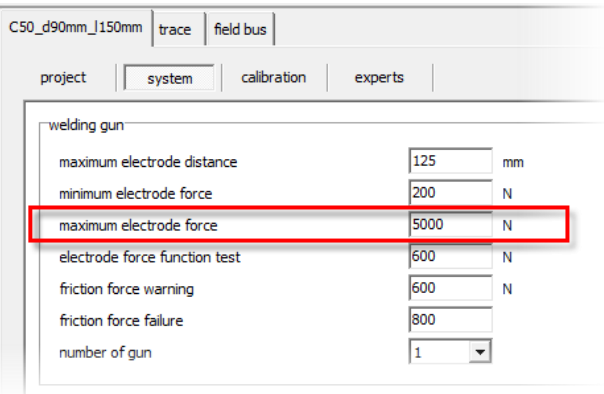
No.	Advice	Comment
4	PC control	 If the PC control is activated, the welding gun can be controlled through the WinSPZ software and placed in operation. The signals from the robot are ignored. The PC control is automatically ended after 60 seconds if the WinSPZ software is separated from the controller without release of the PC control.  If the PC control is deactivated, the welding gun switches back to the applied signals from the fieldbus. Movements can be triggered through this.  In production operation, the welding gun stops immediately after activation of the PC control. This can result in collisions during positioning of the robot!
5	New project	Opens the predefined standard parameter files for the respective welding gun type. For example: C50 with main cylinder diameter 90 mm is the file C50_d90mm.spz 
6	Open project file	 Open SPZ project file

No.	Advice	Comment
7	Save project file as	 Save parameter as SPZ project file 
8	Save project as	Selection of a filename for the project 

8.4 Load base parameters

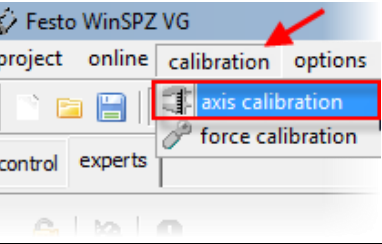
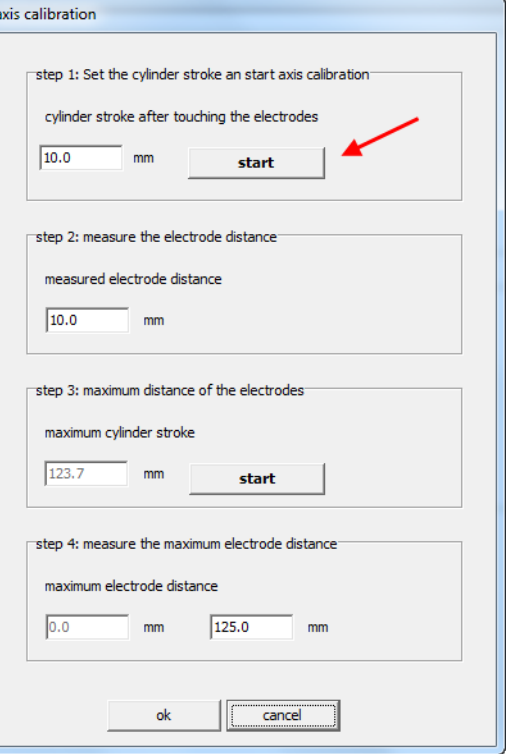
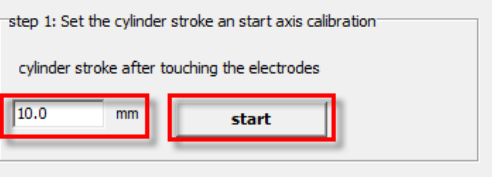
The base parameters contain optimised parameters for

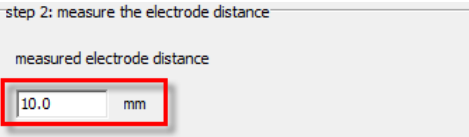
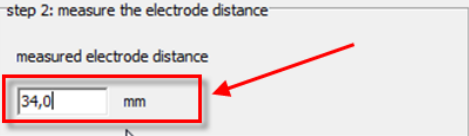
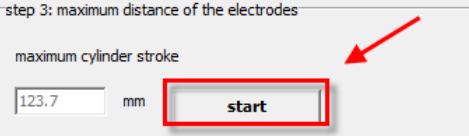
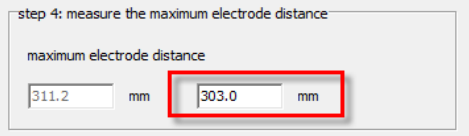
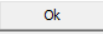
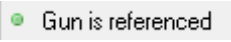
- position control
- force control
- 7 axis operation
- fitting main cylinder
- parameters for electrode caps

No.	Action	Comment
1	Open new project	 The folder (..)default parameter) with the standard parameter files is opened.
2	Select parameter file xxx.spz	Standard parameter sets should be stored for all welding gun types. A predefined data record is present, dependent on the welding gun type used and diameter of the main cylinder. For example: For an X100 welding gun with main cylinder diameter of 140 mm and stroke of 150 mm, the parameter set “X100_d140mm_l150mm.spz” must be used. 
3	Set maximum electrode force	Take the maximum electrode force from the data sheet or rating plate of the welding gun. 
4	Load data into the controller	

8.5 Perform axis calibration

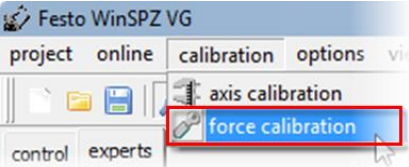
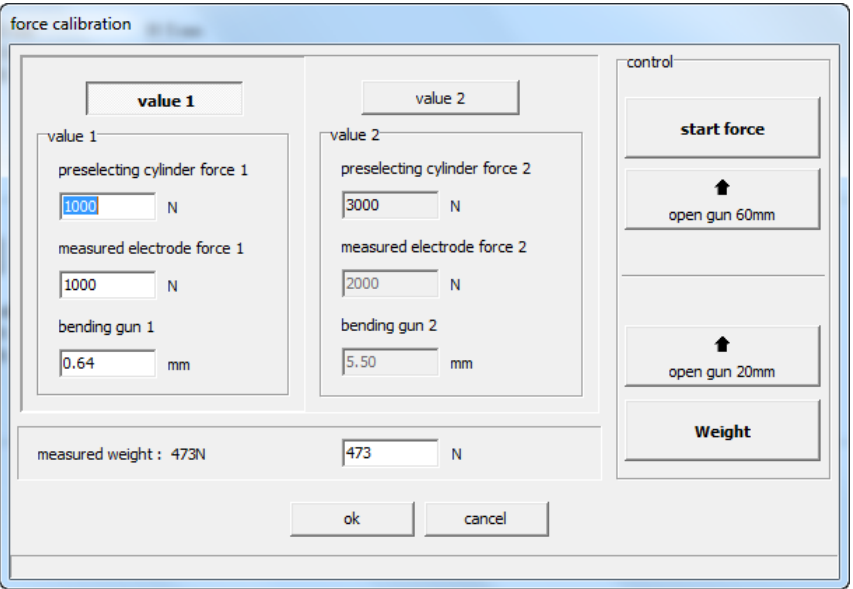
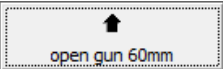
With the axis calibration, the zero position, translation and maximum opening of the welding gun are determined.

No.	Action	Comment
1	Mount new electrode caps	 Perform axis calibration only with new caps and electrode shafts. If machined caps are used, an incorrect zero point is stored in the controller.
2	Connect US2 load voltage supply	 US2 ein
3	Activate PC control	
4	Open axis calibration dialog box	
5	Axis calibration dialog box appears	
6	Set cylinder stroke and actuate Start	The welding gun travels to the electrode caps and opens to the set cylinder stroke. C-gun 10 mm X-gun 10 mm 

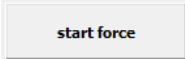
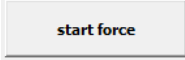
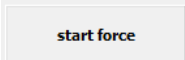
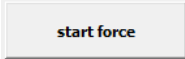
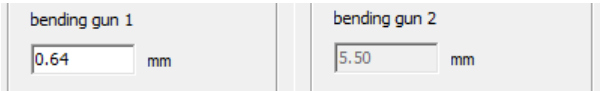
No.	Action	Comment									
7	Enter electrode stroke	C-gun: translation = 1.0 Enter electrode distance 10 mm.  X-gun: translation = electrode stroke / cylinder stroke Measure electrode distance and enter it. 									
8	Open electrodes to the maximum	Welding gun opens to the maximum gun opening.  The maximum cylinder stroke between the electrode and the maximum opening is displayed.									
9	Enter measured electrode distance	C-gun: Measure electrode distance and enter it X-gun: Measure electrode distance and enter it.  Note: In the left field, the calculated maximum welding gun opening is displayed from translation 1.									
10	Close axis calibration with OK	The data are transmitted to the controller and stored in the EEPROM. 									
11	Check values	axis calibration  <table border="1"> <tbody> <tr> <td>cylinder stroke</td> <td>10.0 mm</td> <td>91.5 mm</td> </tr> <tr> <td>electrode stroke</td> <td>34.0 mm</td> <td>303.0 mm</td> </tr> <tr> <td>electrode ratio</td> <td>3.40</td> <td>3.31</td> </tr> </tbody> </table> C-gun: Translation 1 and translation 2 should be equal. X-gun: Translation 2 can be somewhat smaller than translation 1.  Improper calibration can result in collisions.	cylinder stroke	10.0 mm	91.5 mm	electrode stroke	34.0 mm	303.0 mm	electrode ratio	3.40	3.31
cylinder stroke	10.0 mm	91.5 mm									
electrode stroke	34.0 mm	303.0 mm									
electrode ratio	3.40	3.31									
12	Check fieldbus signal “Gun is referenced”	That the welding gun is referenced is displayed to the robot. The welding gun follows the robot only if the signal is one. 									

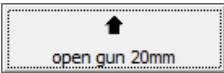
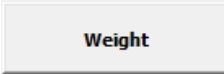
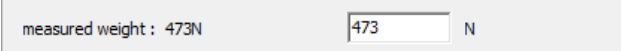
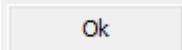
8.6 Perform force calibration

Prepare force calibration

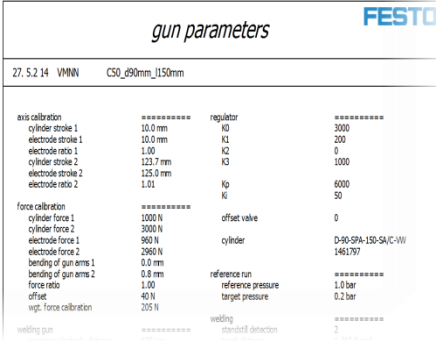
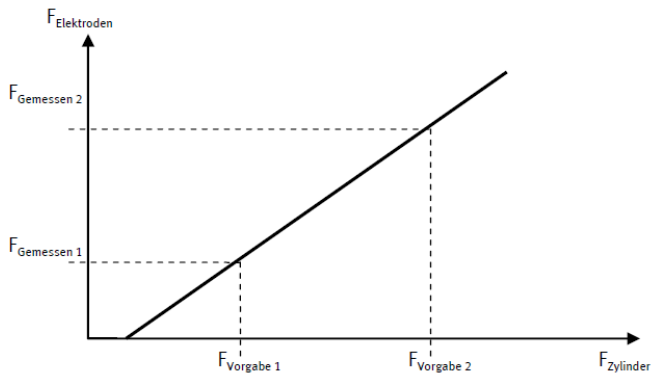
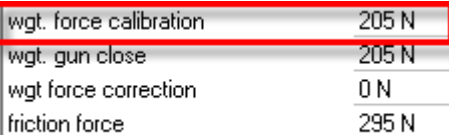
No.	Action	Comment
1	Activate PC control	
2	Preparation	Position welding gun in load-neutral position , so that the cylinder is in a horizontal position. The operating pressure in force calibration should correspond to the later operating pressure in production. Check that all guides and bearings move easily. Perform force build-up several times ($\geq 3x$) and measure the force.
3	Open “Force calibration” dialog box	
4	“Force calibration” dialog box appears	
5	Open gun	Electrode opens to 60 mm. 
6	Determine the force values to be calibrated	Value 1 should be calibrated at 1000 N electrode force. Value 2 should be calibrated at maximum electrode force – 10 %. Note: Hump welding guns For hump welding guns, calibration should be performed near the electrode force for production. Example: 700 N electrode force in production Value 1 = 700 N with electrode force Value 2 = 1200 N with electrode force
7	Perform force calibration	

Perform force calibration

No.	Action	Comment
1	Perform force calibration	Perform force calibration for value 1 and value 2.
1	Place dynamometer	
2	Select value 1 or 2	
3	Specify cylinder force	To achieve the desired electrode force, the correct cylinder force must be specified. Cylinder force 1 = 1000 N * Force translation of the welding gun Cylinder force 2 = Max. electrode force * Force translation of the welding gun 
4	Perform force build-up	
5	Enter electrode force	Enter measured electrode force. 
6	Open gun	
7	Determine bending gun	The negative electrode position (less than zero position) is automatically entered as deflection or arching of the gun arms.
8	Remove dynamometer	Measure deflection of the gun arms for value 1 and value 2. 
9	Perform force build-up	
10	Open gun	
11	Determine bending of gun	The negative electrode position is automatically entered as bending of gun. (Wait approx. 3 seconds) 
12	Determine weight and friction force of the moving gun arm	A reference force for friction and arm weight is determined in the force calibration for automatic, position-dependent force equalization. In the force build-up, the position-dependent arm weight and friction force are automatically corrected. The function "Automatic correction of weight" can be deactivated through input of 0 N measured weight force or in the Experts dialog box and Welding category.

13	Open gun to 20 mm	Electrode opens to 20 mm. 
14	Close gun	Actuate the “Weight” button several times. 
15		The measured weight force is entered. 
16	End calibration of force with OK	The data are transmitted to the controller and stored in the EEPROM. 

General information on force calibration

No.	Advice	Comment
1	Check values 	The ratio of cylinder force to electrode force should correspond approximately to the axis calibration translation. This can be checked very easily in the printout of the welding gun parameters. axis calibration ===== cylinder stroke 1 10.0 mm electrode stroke 1 10.0 mm electrode ratio 1 1.00 cylinder stroke 2 123.7 mm electrode stroke 2 125.0 mm electrode ratio 2 1.01 force calibration ===== cylinder force 1 1000 N cylinder force 2 3000 N electrode force 1 960 N electrode force 2 2960 N bending of gun arms 1 0.0 mm bending of gun arms 2 0.8 mm force ratio 1.00 offset 40 N wgt. force calibration 205 N
2	Force scaling is determined for 2 values	
3	The measured weight force is displayed in the Current Values window	

8.7 Set parameters for welding gun

welding gun		
maximum electrode distance	125	mm
minimum electrode force	200	N
maximum electrode force	3000	N
electrode force function test	600	N
friction force warning	600	N
friction force failure	800	
number of gun	1	

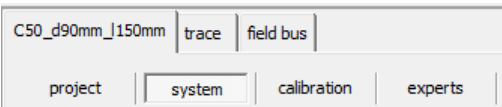
Figure 10: Parameters for welding gun

No.	Parameter	Comment
	Open System menu	
1	Maximum opening dimension [mm]	Maximum gun opening can limit the maximum opening of the gun as a software limit switch. The value is determined automatically in axis calibration.
2	Minimum electrode force [N]	Minimum permissible electrode force
3	Maximum electrode force [N]	Maximum permissible electrode force of the gun
4	Electrode force function test [N]	Reduced electrode force for performance test without component (simulation mode)
5	Friction force warning [N]	Limit until the warning message "Increase in friction" is detected
6	Friction force failure [N]	Limit until the fault signal "Gun moving hard" is output on the fieldbus
7	Number of gun	Number of gun for docking guns. If the gun number does not match the number specified by the robot on the fieldbus, the bit "Invalid gun code" is output.
8	Actuate download	Parameters become active after they have been transmitted to the controller

8.8 Set parameters for electrode caps

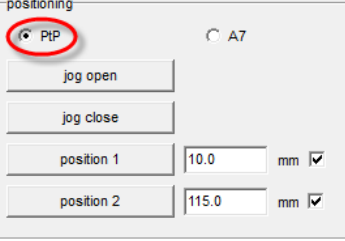
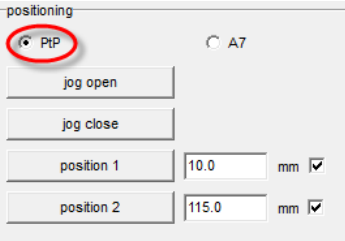
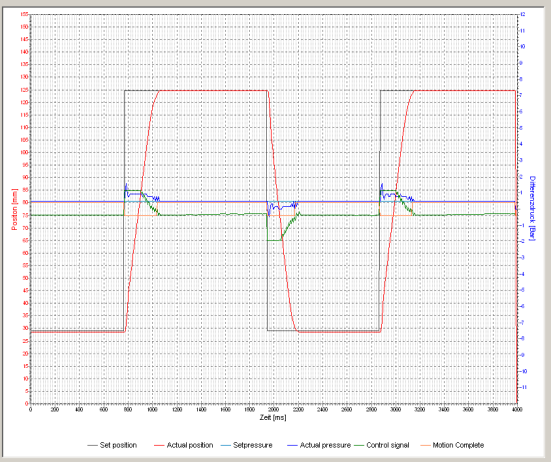
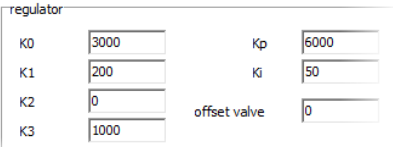
electrode caps		
cap life quantity	6.5	mm
end measure E-cap fixed arm	10.0	mm
end measure e-cap moving arm	10.0	mm
tolerance geometry check	6.0	mm
tolerance set caps	7.0	mm
max. increase per milling cycle	0.3	mm
max. decrease per milling cycle	2.0	mm

Figure 11: Parameters for electrode caps

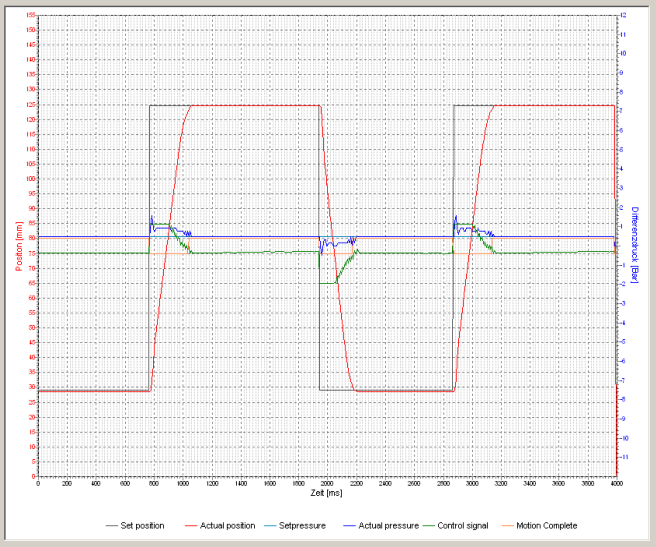
No.	Parameter	Comment
	Open System menu	
1	Cap life quantity [mm]	Maximum size that can be milled from a single electrode cap. The cap life is checked after each ☐ “measurement after milling”
2	End measure e-cap fixed arm / moving arm [mm]	Measure the old caps after exchange and enter the height of the cap. If the caps were milled unsymmetrically, this ratio is taken into account in the TCP reset value to the robot. If the caps are milled evenly, the same dimension must be entered for both the fixed and the moving e-arm.
3	Tolerance, geometry check [mm]	Maximum deviation from the zero position allowed for new caps. After each measure before milling, the cap position is compared to the zero position in axis calibration. If the tolerance is exceeded, a geometry error is reported.
4	Tolerance set caps [mm]	At each setting stroke, the cap position is compared to the zero position. The cap loss is checked. If the tolerance is exceeded, a geometry error is reported.
5	Max. increase per milling cycle [mm]	The current cap position is compared to the last cap position after each measurement after milling. For the entry of 0 mm the caps must be shorter after each milling (chip monitoring). If the tolerance is exceeded “milling n-IO” is reported.
6	Max. decrease per milling cycle [mm]	The caps can be milled off per milling cycle by up to this dimension. If the tolerance is exceeded, an error during milling is reported.
7	Actuate download	Parameters become active after they have been transmitted to the controller 

8.9 Set controller parameters / check positioning behaviour

Optimise controller parameters

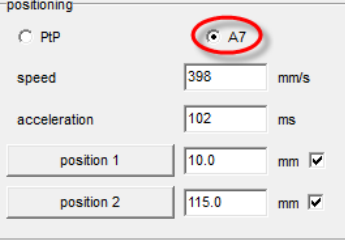
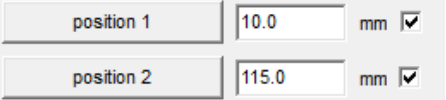
No.	Action	Comment
1	Enable PC control	
2	Switch gun to positioning mode PtP (point-to-point)	
3	Activate trace	Trace is started after each change of setpoint position 
3	Change setpoint position	Alternate actuation of buttons for position 1 and 2 The gun moves at maximum speed to the desired position. 
4	Evaluate trace	The cylinder position (red) should move to the setpoint position (black) without overshooting, if possible. 
5	Optimise controller setting	
6	Actuate download	Parameters become active after they have been transmitted to the controller 
7	Go to step 3	Change settings until the result is satisfactory.

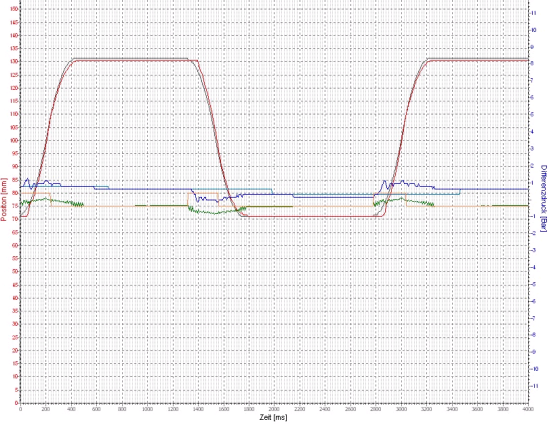
Parameters for position control

No.	Advice	Comment	
1	Typical setting values for C-guns	Cylinder diameter [mm] 90 100 125	Parameter K0:3000 K1:200 K2:0 K3:1000
1	Typical setting values for X-guns	Cylinder diameter [mm] 100 125 140 160	Parameter K0:3500 K1:400 K2:0 K3:2000 K0:3500 K1:300 K2:0 K3:3000
2	Parameter K0	Set proportional amplifier K0 so that positioning takes place without overshooting.	
3	Parameter K1	Set cushioning for speed K1 if the gun tends to overshooting.	
4	Parameter K2	Increase acceleration cushioning K2 to compensate for friction effects.	
5	Parameter K3	Increase cushioning for pressure change K3 if the gun tends to oscillate when it reaches the target position or when moving slowly.	
6	Check positioning behaviour. Setpoint position (black) Check positioning signal of proportional valve (green) and actual position of cylinder (red)		

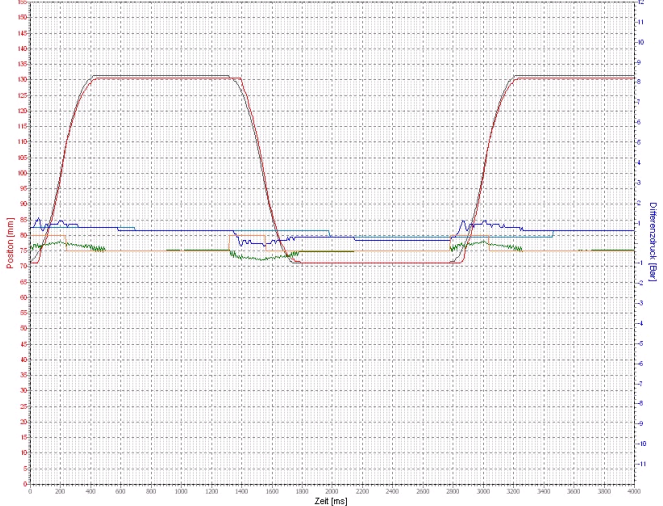
8.10 Set and check parameters for 7 axis operation

In 7 axis operation, an internal ramp generator is placed upstream for simulation in the controller. As a result, there is the possibility to check the controller behaviour without robots and optimise the parameters for the pilot control. In addition to the setpoint position, the possibility exists to specify the speed and acceleration for the internal ramp generator.

No.	Action	Comment
1	Enable PC control	
2	Switch gun to 7 axis operation	 
3	Activate trace	Trace is started after each change of the setpoint position. 
4	Set speed and acceleration Change setpoint position	For example: Cylinder diameter 90 mm Speed 400 mm/s Acceleration 100 ms Cylinder diameter 140 mm Speed 180 mm/s Acceleration 150 ms Alternate actuation of position 1 and position 2 

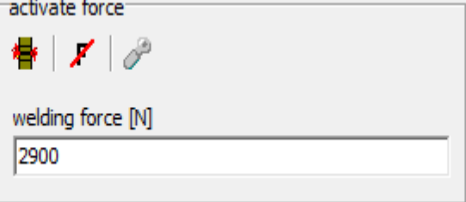
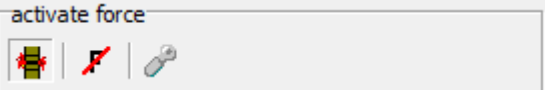
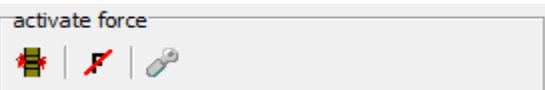
No.	Action	Comment
5	Evaluate trace	 Black, cylinder setpoint position Red, cylinder actual position Target: Actual position should follow setpoint position with little space Check at various speeds. Gun should position itself without oscillation of the gun arms even at slow speed.
6	Set following error equalization	
7	Actuate download	Parameters become active after they have been transmitted to the controller. 
8	Go to step 4	Change settings until the result is satisfactory.

Parameters for 7 axis operation

No.	Advice	Comment
1	Typical setting values	Kv static (50 -100) Ka static (0 -20) Kv dynamic (50 -100)
2	Parameter Kv static	Static pilot control of the setpoint speed.
3	Parameter Ka static	Static pilot control of the setpoint acceleration (Parameter is not used).
4	Parameter Kv dynamic	Integrating error correction.
5	Setting Kv static	Optimise Kv static Ka static = 0 Kv dynamic = 0 Typical values for C-guns Kv static (50 -100) Typical values for X-guns Kv static (70 -150)
6	Check positioning behaviour Check setpoint position (black) and actual position of cylinder (red)	Set Kv static so that there is a small lead of the setpoint position to the actual position. 
7	Setting Kv dynamic	Set Kv dynamic to 100.
8	Check positioning behaviour	

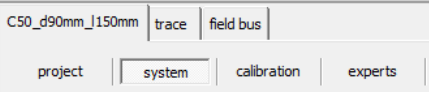
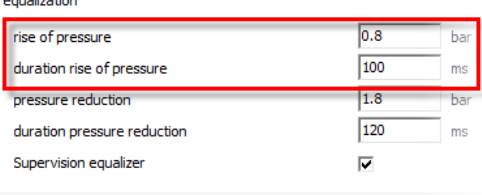
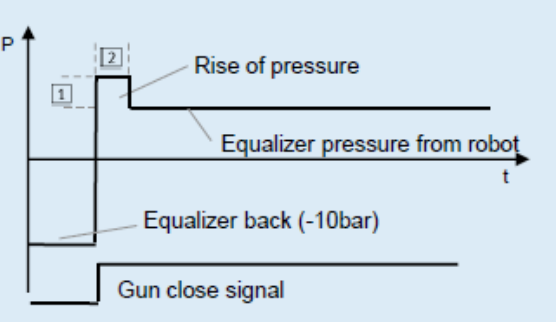
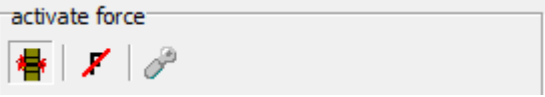
8.11 Set equalizer

Preparation to set equalizer

No.	Action	Comment
1	Enable PC control	
2	Switch gun to positioning mode	
3	Open electrodes to 15 mm	set position [mm] 124.1
4	Specify welding force	Set desired welding force activate force  welding force [N] 2900
5	Specify compensating pressure	Simulate setting value from the robot equalization equalizer pressure [Bar] -10.0
6	Activate force build-up	Gun closes Equalizer is activated with equalizer pressure [bar] activate force 
7	Deactivate force build-up	Gun opens to setpoint position Equalizer travels to initial position activate force 
8	Search for optimal equalization pressure	Adapt equalization pressure until gun is floating Repeat steps 5 – 7

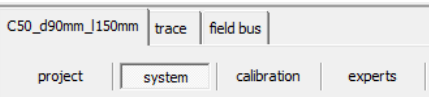
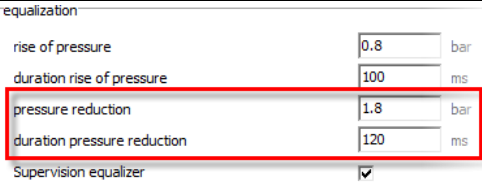
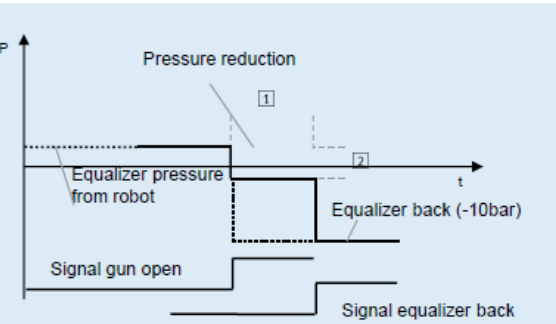
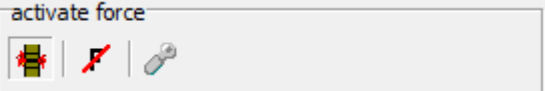
Set rise of pressure

With the rise of pressure, a pressure pulse is generated to overcome friction. The stuck electrode should thereby move to the component.

No.	Action	Comment
1	Open System menu	
2	Adjust parameter for rise of pressure	
3	Actuate download	
4	1 Rise of pressure [bar] 2 Duration of rise of pressure [ms]	
5	1 Rise of pressure [bar] 2 Duration of rise of pressure [ms]	Adjust values until the stuck electrode moves to the component. Typical values: Rise of pressure: 0.5-1.5 bar Duration: 50 – 150 ms
6	Actuate force build-up	
7	Adapt settings	Repeat step 2 until an optimal result is achieved.

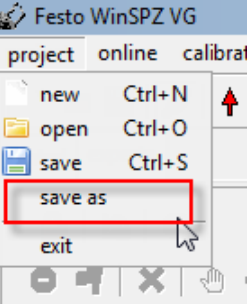
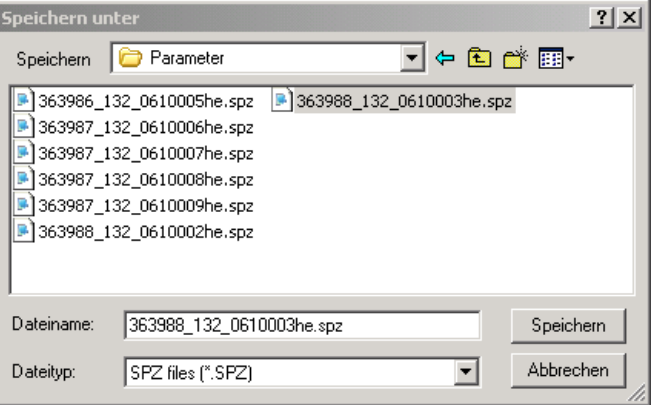
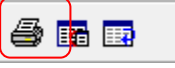
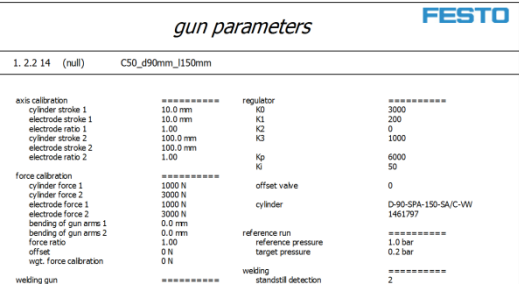
Adjust pressure reduction

With pressure reduction, the return to the initial setting during opening of the gun can be influenced. An intermediate step for pressure reduction is thereby parameterized. As soon as the equalizer pressure is less than -4.5 bar, the signal “Equalizer back” is sent to the robot. Optionally, the initial position of the gun can also be sensed with a limit switch at input 2.

No.	Action	Comment
1	Open System menu	
2	Set parameters for pressure reduction	
3	Prepare parameters for pressure reduction	
4	Adapt pressure reduction	Increase pressure reduction until the gun travels gently into the initial setting. Typical values: 1.0-3.0 bar
5	Actuate download	
6	1 Duration of rise of pressure [ms] 2 Pressure reduction [bar]	
7	Set duration of pressure reduction	Reduce the duration of pressure reduction so that the gun immediately sets the pressure to 10 bar once the initial setting is reached. Typical value: 150 – 200 ms Note: The signal “Equalizer back” is reported to the robot only after this time has passed.
8	Actuate force build-up	
9	Adapt settings	Repeat step 4 until an optimal result is achieved.

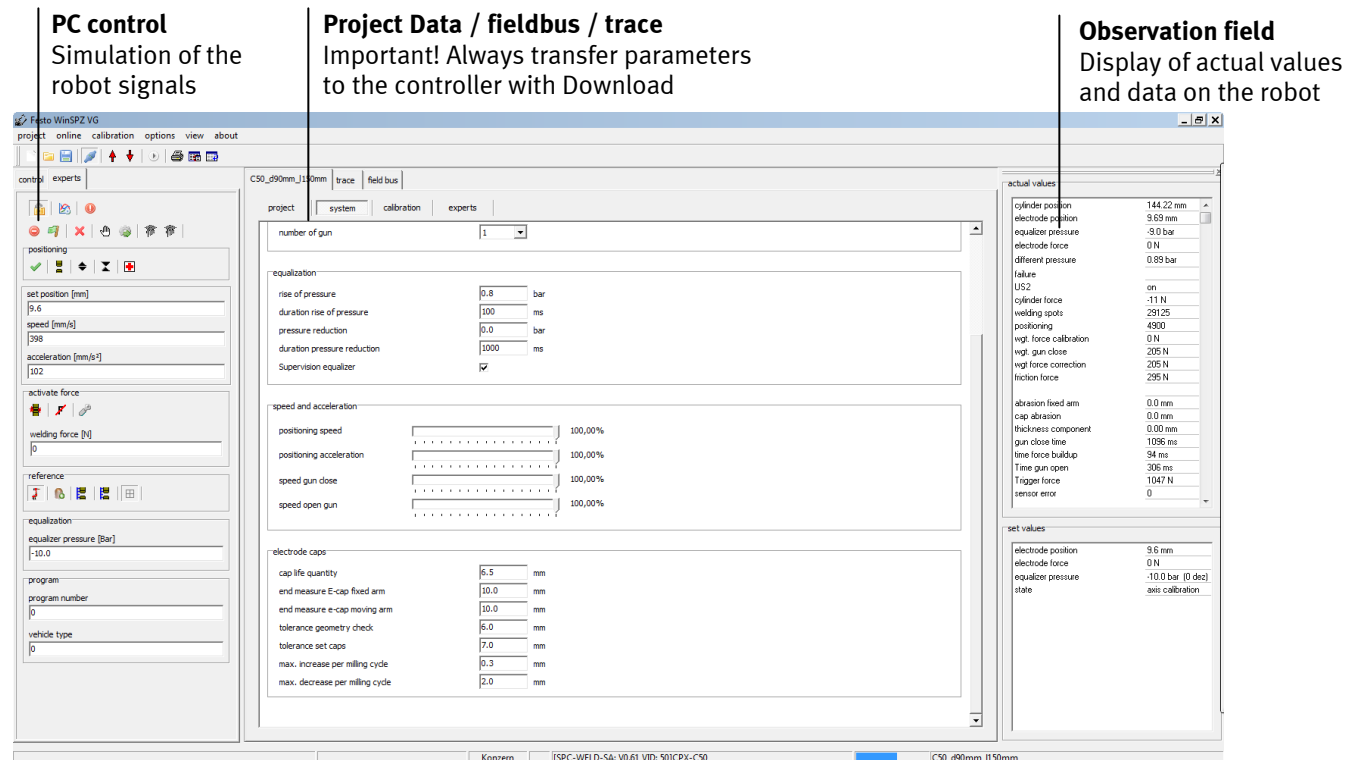
8.12 Save project data

Save parameter as project file

No.	Action	Comment
1	Select upload Data are read out of the controller	
2	Save project as	
3	Enter filename	
4	Print project data	 

9 Software WinSPZ

9.1 Overview of user interface, WinSPZ



9.2 System requirements

Festo WinSPZ is supplied as a setup file (e.g. Festo WinSPZ VG Setup 1.0.3.5.exe).

The software does not write any data into the Windows Registry and requires no database drivers or other Windows resources.

The WinSPZ software can also be copied to a memory stick and started directly from there.

For operation of the WinSPZ software, you need a Pentium-compatible PC with the following minimum system requirements.

- Microsoft Windows 2000 or XP or Windows 7,
- Recommended main memory: at least 1 GB,
- 20 MB of free space on hard disk,
- CD-ROM drive,
- A free USB port and a USB adapter corresponding to chapter 10.7.1 or alternatively one free Ethernet port for connection of the controller (only for online mode).

9.3 Installation

To install the software package WinSPZ on your hard disk, carry out the following steps.

1. Start Microsoft Windows.
2. Place the installation CD in the CD-ROM drive.
3. Go to the Start menu and select the command [Run].
4. In the “Run” dialog box, enter the drive letter and the name of the installation program (e.g. D:\SETUP.EXE).
5. Start the installation program with “OK”
6. Follow the instructions in the installation program.

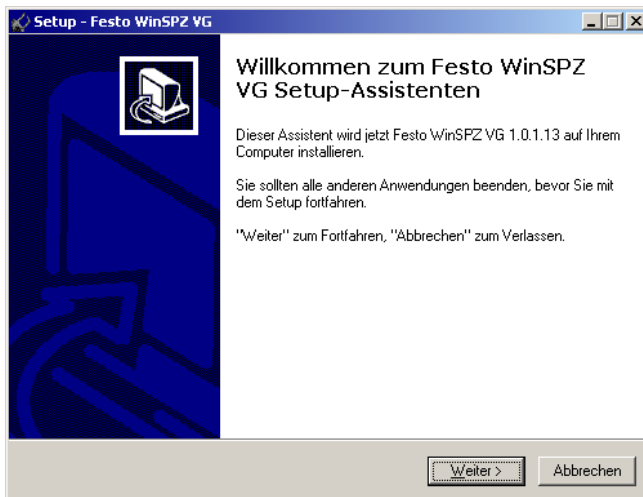


Fig. 9.1: WinSPZ installation program

7. Select “Festo WinSPZ, start” and click on “Finish”

9.4 Deinstallation



WinSPZ can be deinstalled using the “Add or Remove Programs” function in the Windows Control Panel. Proceed here as described in your Windows manual.

9.5 Function range

Main features

- WinSPZ is used for setting up welding gun systems:
 - Project engineering
 - Commissioning
 - Simulation
 - Performance test
 - Diagnostics
- Different controllers:
 - Type 03 / CPX series
- Different fieldbus systems:
 - ProfiNet
 - InterBus
 - ProfiBus
 - DeviceNet
- Different fieldbus protocols:
 - Audi B8 protocol, VW Group standard
- Multilingual:
 - German, English, Portuguese, Spanish, Chinese
- Different interfaces:
 - RS 232
 - Ethernet via system PC

Function range

- Setting of the gun parameters
- Parameterisation of the pressure and position controller
- Performance test and simulation via PC controller
- Trace (graphic representation of setpoint values and actual values)
- Online display of fieldbus signals
- Diagnostics: representation of the current status (position, reference position, orifice,)
- Axis calibration
- Force scaling
- Printing of the gun parameters
- Loading and saving of the project data
- Firmware update

9.6 Program summary

The WinSPZ program contains the following operation and information elements (see following figure).

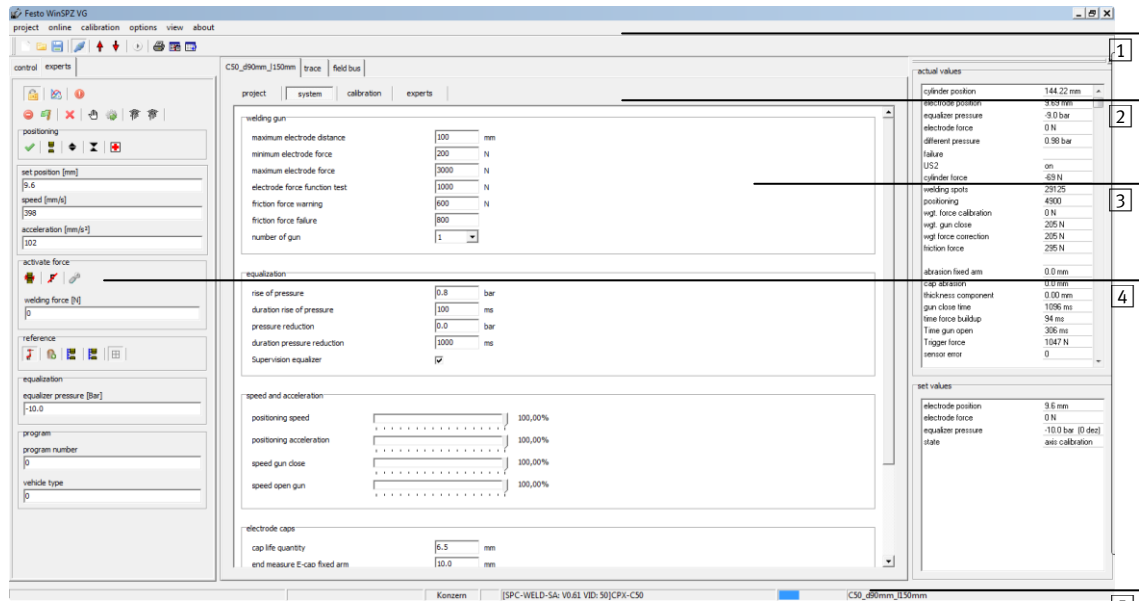


Fig. 9.2: Program window WinSPZ

No.	Area/panel	Contents
1	Main menu and toolbar	In the main menu, you will find all the commands for general control of the program. The toolbar contains buttons for the most frequently used commands.
2	“Project”	This panel contains all information about the currently loaded project, displaying all components and parameters of the welding gun drive in that project.
3	“Actual values/setpoint values”	This panel contains not only the setpoint values but also the actual values, as long as the connection to the controller of the welding gun drive is active.
4	“Control”	This panel can be used to control the welding gun drive directly.
5	“Status line”	Display of the firmware version of the connected controller. Progress bar shows the communication to the controller.

Table 9.1: Elements in the WinSPZ program window

9.6.1 Menu commands

[Project] menu

The [Project] menu contains the commands for generating, saving and reloading projects, which contain all the relevant configuration, parameterisation and control data.

Command/button		Function
[New]		Opens the directory with the standard parameter project files. Installation folder\default parameters
[Open]		Opens a saved project file; if there is an existing connection: permits system data synchronization and transfer of the file contents to the controller of the welding gun drive.
[Save]		Saves the current project file. When the project file is saved for the first time, the “Save file as” dialog box will appear so that you can enter a name.

Table 9.2: [Project] menu

[Online] menu

The commands in this menu are used for reception of the online connection and for uploading and downloading projects.

Command/button		Function
[Connect]	 	Establishes the connection to the controller of the welding gun system. Clicking this menu command again terminates the active connection.
[Upload]		Reads the project data from the controller. The data are transmitted from the controller to the PC.
[Download]		Writes the project data to the controller. The data are transmitted from the PC to the controller and stored in the EEPROM.

Table 9.3: [Online] menu

[Calibration] menu

The commands in this menu are used for calibrating the welding gun system.

Command/button		Function
[Axis calibration]		Determines zero position and translation of the electrodes.
[Force calibration]		Scaling of the electrode force

Table 9.4: [Calibration] menu

[Options] menu

Command	Function
[Drives]	Opens the drive database.
[Firmware]	Opens the submenu for downloading the firmware.
[Language]	Setting the programming language.

Table 9.5: [Options] menu

[View] menu

Command/button	Function
[Control]	Opens / closes the “Control” panel.
[Setpoint/actual values]	Opens/closes the “Setpoint/actual values”.

Table 9.6: [View] menu

[About] menu

Command	Function
[Info]	Version information about the software.

Table 9.7: [About] menu

9.6.2 Toolbar of the program window

- | | |
|--|---|
| 1 Open project file for a welding gun system | 5 Transfer parameters from the controller to the PC |
| 2 Open project file | 6 Transfer parameters from the PC to the controller |
| 3 Save project file | 7 Starts the graphic recording (trace) |
| 4 Establish or disconnect connection to the controller of the welding gun system – online mode | 8 Printable view of all project data |
| | 9 Save project data as CSV file |
| | 10 Create project data from CSV file |

Fig. 9.3: Toolbar for quick access to important and frequently needed functions

9.6.3 Show and hide the panels

The panels “Control” and “Setpoint/actual values” can be shown and hidden as needed using the [View] menu.

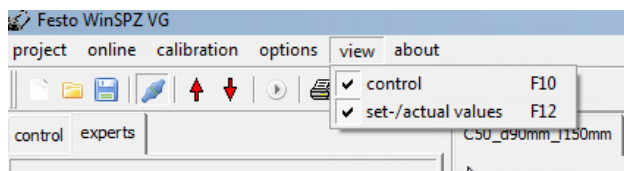


Fig. 9.4: “View” menu commands

9.7 Online mode

For communication between WinSPZ and the controller, you need to activate the online mode. Here, a connection is built up between the PC and the controller via a serial interface or, if a ProfiNet fieldbus node is used, via an Ethernet connection.

9.7.1 Connection PC - CPX controller with RS232 USB adapter

To connect the service interface on the CPX node to the PC, use the adapter (NEFC-M12G5-0.3-U1G5) and also a conventional USB cable with mini USB connector.

Information on installing the adapter and on the COM1 interface used can be found in the documentation for the adapter.

- 1 Adapter
- 2 USB cable

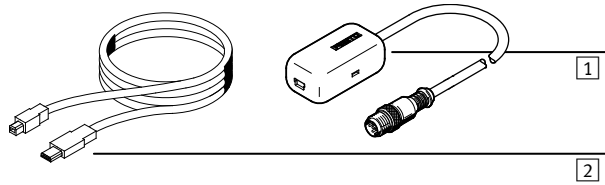


Fig. 9.5: Adapter for connection to the PC



Note

The service interface (V.24) of the CPX terminal is not galvanically isolated. It is not intended for permanent connection to PC systems or as a control interface.

- Remove the adapter during continuous operation.



Follow the instructions for operation and for installation of the required drivers on the accompanying CD-ROM or on the Internet:

→ www.festo.com → Support/Downloads → Enter search term: FMT

After commissioning, seal the service interface of the CPX node with the protective cap supplied.

9.7.2 Enable or deactivate online mode

Once you have connected the PC and controller, go to the [Online] menu and select [Connect], or click on “Establish connection” on the toolbar.

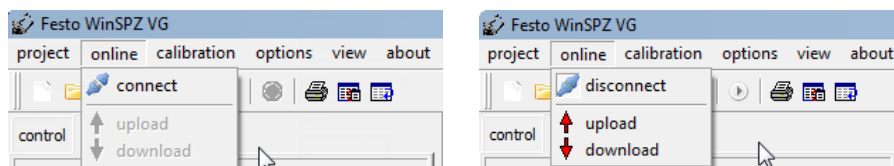


Fig. 9.6: Online mode

The connection assistant is opened.

9.7.3 Connection assistant

The connection assistant searches for all controllers connected to the PC. Select the controller you wish to connect to and then click on [Connect].

If “Perform upload” is activated, the project data is uploaded during connection set-up.

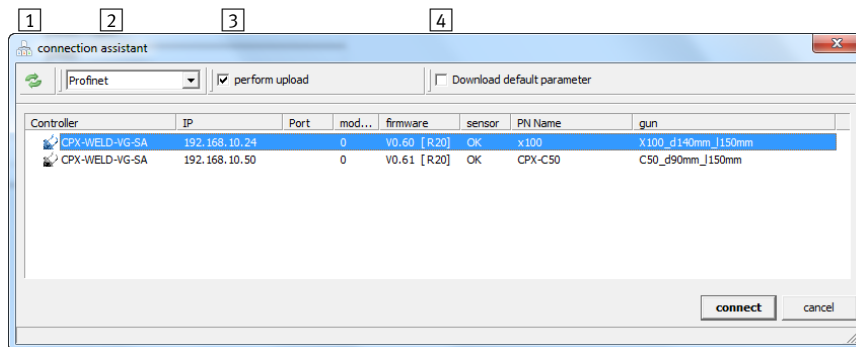


Fig. 9.7: Connection assistant

- 1 Update: Load the network again
- 2 Protocol: Select the communication interface (ProfiNet or serial)
- 3 Perform upload: Triggers an upload when connecting
- 4 Download default parameter: Load factory parameters

The connection assistant shows the following data for the connected controller:

- IP address (for Profinet controllers only),
- COM port (only if “Serial” protocol was selected),
- Module number, (slot in the CPX node)
- Firmware Version [ProfiNet Revision], (example firmware V0.60 Profinet Revision R20)
- Sensor, (Status of the position sensor OK, Warn, Error)
- ProfiNet name,
- Gun. (Freely selectable gun name)

9.7.4 “Drives” database

This database stores all specifications for the main cylinder.

- Open the “Drives” dialog box through the menu [Options][Drives].

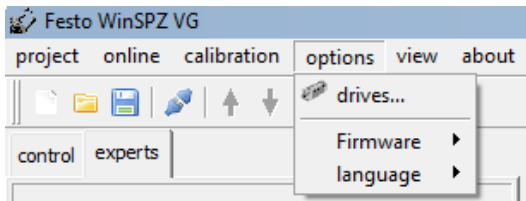


Fig. 9.8: [Drives] menu command

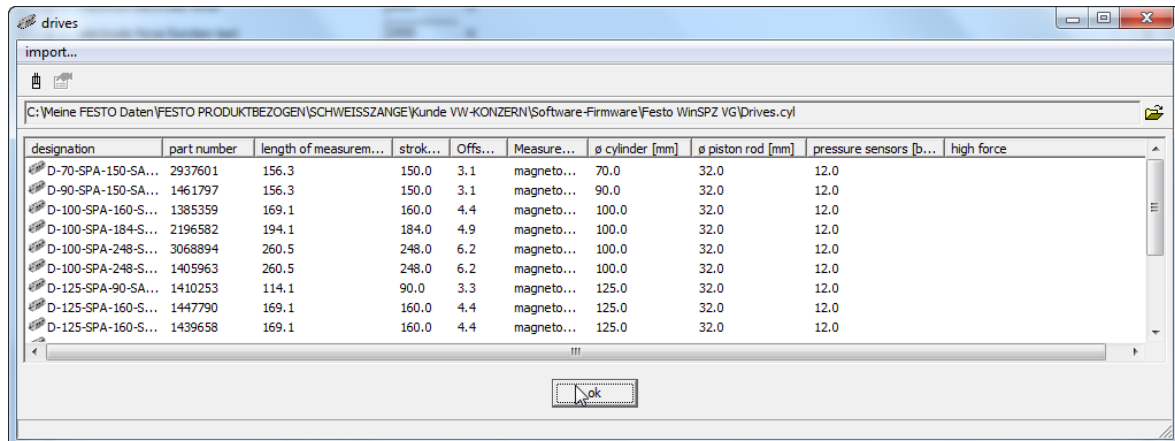


Fig. 9.9: “Drives” database

- 1 Path where the drives database is saved

- Double-click an existing drive to correct its characteristics.

Command/button		Function
[Create new]		Opens a dialog for entering the characteristics of a new drive.
[Open]		Opens the dialog of a selected drive.
[Ok]		Applies a drive from the database to the project.

Table 9.8: Parameterisation of the drives

Create a new drive



- In the “Drives” dialog box, select the “Create new” button.

You will see a dialog box where you can enter all the properties of the new drive.

Fig. 9.10: “Drive characteristics” dialog box

Input field	Contents
Designation	Freely selectable designation of the main cylinder.
Part number	Part number of the cylinder (see rating plate).
Measuring system length	Length of the built-in measuring system.
Stroke length	Length of the mechanical cylinder stroke.
Diameter	Diameter of the piston.
Piston rod diameter	Diameter of the piston rod.
Offset	0-point offset of the measuring system.
Measuring system	Type of measuring system (potentiometer or contactless).
Pressure sensors	Measuring range of the built-in pressure sensors.
High force	Indicates whether the cylinder is a high-force cylinder.

Table 9.9: Drive characteristics

Festo provides the database for all available drives.

9.8 Project window

9.8.1 Project

The specifications in the project window are optional and are intended for the purpose of documenting the project. They are only saved in the *.spz file and not in the controller.

The screenshot shows the 'Project' window in the WinSPZ software. At the top, there are three tabs: 'TEST', 'trace', and 'field bus'. Below these, there are four sub-tabs: 'project', 'system', 'calibration', and 'experts'. The 'project' sub-tab is selected, displaying a form with the following fields:

- title**: A text box containing 'TEST'.
- author**: A text box containing 'Mustermann'.
- created**: A date box containing '28.05.2014'.
- remark**: A large text area containing the text 'welding gun for testing'.

Fig.

9.11: Project window – Project

Parameter	Significance
Title	Freely selectable title of the welding gun project.
Author	Your name.
Created by	Date of creation (set by default to the current date).
Comments	Any text can be entered here.

Table 9.10: Project window elements – Project

9.8.2 System dialog

After a change, parameters in the System group must be loaded to the controller using the download function.

Fig. 9.12: Project window - System

Welding gun

Parameter	Unit	Significance
Maximum electrode distance	mm	Maximum mechanical opening width of the electrodes. If the setpoint position is exceeded by the robot, an error is sent to the robot.
Minimum electrode force	N	Limit value of the smallest permissible electrode force.
Maximum electrode force	N	Limit value of the largest permissible electrode force.
Electrode force function test	N	Force values in performance test without component or metal sheet to prevent the electrodes from sparking over.
Friction force warning	N	Measured frictional force at which a warning is generated.
Friction force failure	N	Measured frictional force at which a malfunction is generated.
Number of gun		Gun coding. Reported over the fieldbus.

Table 9.11: Welding gun

Equalizer

The equalizer is activated by the robot with the signal “Close gun” or “Start learn teach position”. For each welding point, the robot sends the suitable binary value for equalising pressure over the fieldbus.

For all other operating modes, the equalizer is in the basic setting. The gun is thereby held in the initial position by means of the compensating cylinder with the max. operating pressure.

The “Equalizer is back” signal shows the robot that the gun is in the initial position and travel by the robot is allowed. The signal is given when the actual pressure in the valve for control of the equalizer reports less than -4.5 bar. Optionally, the initial position of the gun can be monitored in addition through an end-position switch at input 2.

To overcome friction when activating the equalizer, there is the option to generate a start signal. This start signal is switched on when the equalising pressure is activated. The stuck electrode should thereby move gently to the component.

Parameter	Unit	Significance
Rise of pressure	bar	Height of pressure impulse at activation of the equalising function.
Duration of rise of pressure	ms	Duration of pressure impulse at activation of the equalising function.

Table 9.12: Equalizer parameter

With the “open gun” signal, the equalizer travels back into the initial position. To allow the welding gun to run gently into the initial positioning, a step can be programmed as pressure reduction. After expiration of the time, the operating pressure is switched back on for the equalizer.

Parameter	Unit	Significance
Pressure reduction	bar	Height of pressure reduction.
Duration of pressure reduction	ms	Duration of pressure reduction. After expiration of this time, the full operating pressure is switched on.

Table 9.13: Equalizer parameter

Parameter	Unit	Significance
Monitoring equalizer		Monitoring of the equalising function can be deactivated. This can take place for guns without equalising valve.

Table 9.14: Equalizer parameter

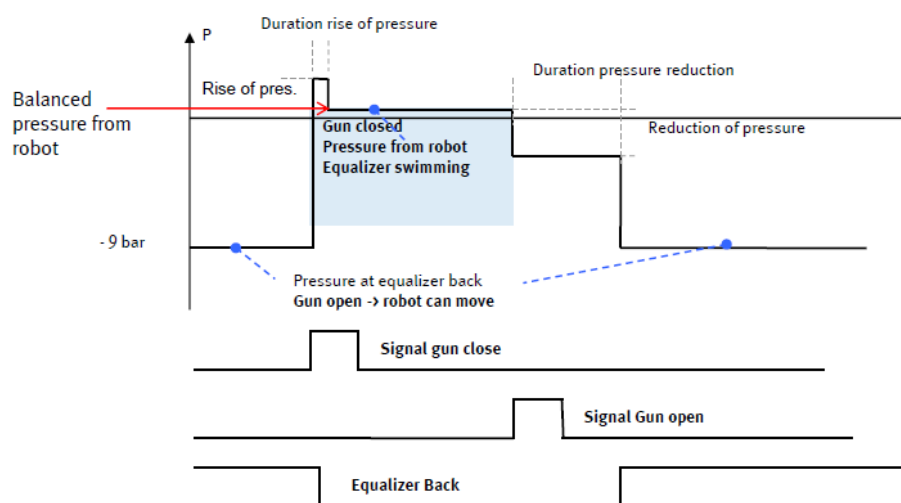


Fig. 9.13: Equalizer behaviour

Speed and acceleration

Parameters for the speed and acceleration of the main cylinder can be set in order to reduce torques when moving large loads, for example.

Parameter	Significance
Positioning speed	Speed of movements of the main cylinder (does not apply for 7 axis operation).
Positioning acceleration	Acceleration of the movements of the main cylinder (does not apply for 7 axis operation).
Speed gun close	Speed when travelling to the component
Speed open gun	Speed during opening of the gun

Table 9.15: Speed and acceleration

Example A

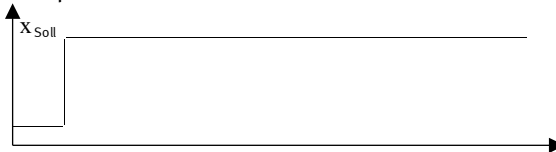


Fig. 9.14: Maximum speed and acceleration ($v = 100\%$; $a = 100\%$)

Example B

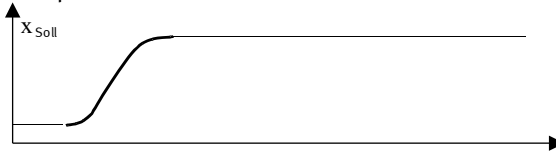


Fig. 9.15: Reduced speed and acceleration ($v < 100\%$; $a < 100\%$)

Electrode caps

Parameter	Unit	Significance
Cap life quantity	mm	Maximum life per electrode cap. The cap life is monitored after each milling. If the cap life is fallen below, an error is reported to the robot.
End measure e-cap fixed arm	mm	Height of the old electrode cap of the fixed electrode arm after cap exchange. Unsymmetrical milling of the caps is taken into account. The wear dimension of the electrode cap of the fixed electrode arm is sent to the robot.
End measure e-cap moving arm	mm	Height of the old electrode cap of the moving electrode arm after cap exchange.
Tolerance geometry check	mm	After each cap dressing, the electrode caps with the signal “measure before milling” are measured and compared to the zero position from the axis calibration. As a result, shaft reduction or setting of incorrect caps can be monitored. If the tolerance is exceeded, an error is reported to the robot.
Tolerance set caps	mm	At each cap dressing, the caps are pressed onto the shaft through a setting stroke and thereby measured against the zero position. If the tolerance is exceeded, an error is reported to the robot.
Max. increase per milling cycle	mm	After each milling, the caps are measured through the signal “measurement after milling”. The measured position is compared to the last position saved and checked for this tolerance. With input of 0 mm, the caps must be shorter after each milling.
Max. decrease per milling cycle	mm	The electrode caps can be milled off per milling cycle by a maximum of this dimension.

Table 9.16: Parameters for electrode caps

9.8.3 Calibration dialog

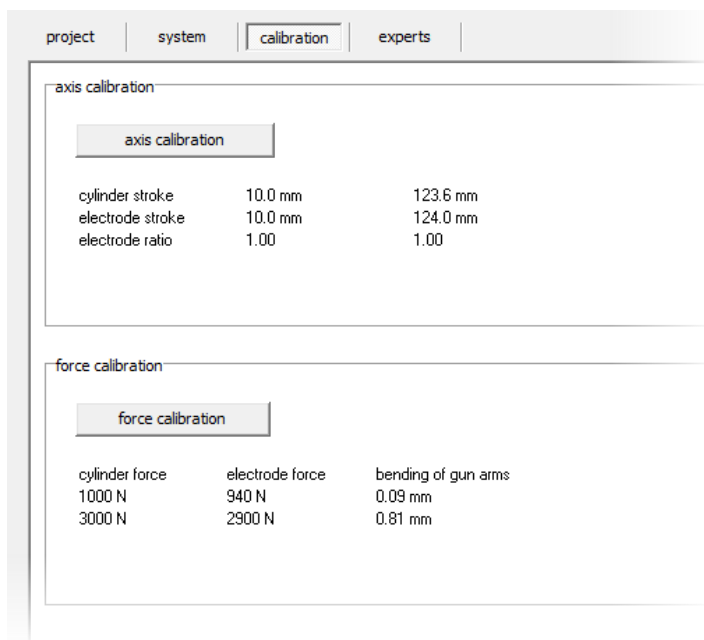


Fig. 9.16: Calibration

Axis calibration

Overview of the axis calibration parameters

Parameter	Unit	Significance
Axis calibration button	–	Opens the “Axis calibration” dialog box.
Cylinder stroke 1	mm	Specified cylinder stroke 1
Cylinder stroke 2	mm	Cylinder stroke for maximum gun opening
Electrode stroke 1	mm	Measured electrode stroke with 10 mm cylinder stroke
Electrode stroke 2	mm	Measured electrode stroke with transmission ratio determined at maximum gun opening
Electrode ratio 1	–	Transmission ratio determined with 10 mm cylinder stroke
Electrode ratio 2		Transmission ratio determined at maximum gun opening

Table 9.17: Axis calibration

Force calibration

Overview of the force calibration parameters

Parameter	Unit	Significance
Force calibration button	–	Opens the “Force calibration” dialog box.
Cylinder force 1, 2	N	The cylinder force specified in the force calibration.
Electrode force 1, 2	N	Measured electrode force.
Bending of gun arms 1, 2	mm	Determined bending of the gun arms.

Table 9.18: Force calibration

9.8.4 Experts dialog

Fig. 9.17: Experts dialog

Regulator

Parameter	Significance
K0, K1, K2, K3	Amplification gains of the position controller.
Kp and Ki	Amplification gains of the force and pressure regulator.
Valve offset	Offset for valve.
Cylinders	Main drive used (from the database). Includes information on the position transmitter, stroke and diameter of the main drive.

Table 9.19: Regulator

Reference run

The gun closes with slow speed until the reference pressure is reached. The pressure is reduced until the target pressure is reached. The position of the electrodes is measured when the target pressure is reached. Then the gun opens to the setpoint position of the robot.



For guns with heavy electrode arms or high friction, the target pressure and reference pressure can be adjusted correspondingly.

Parameter	Unit	Significance
Reference pressure	bar	When this pressure value is reached, it is recognised that the gun is closed.
Target pressure	bar	The differential pressure is reduced to this value. When the pressure is reached, the position in the main cylinder is measured.

Table 9.20: Reference run

- 1 Reference run
- 2 Pressure profile in the main cylinder
- 3 Break-away pressure
- 4 Reference pressure
- 5 Gun position
- 6 Target pressure

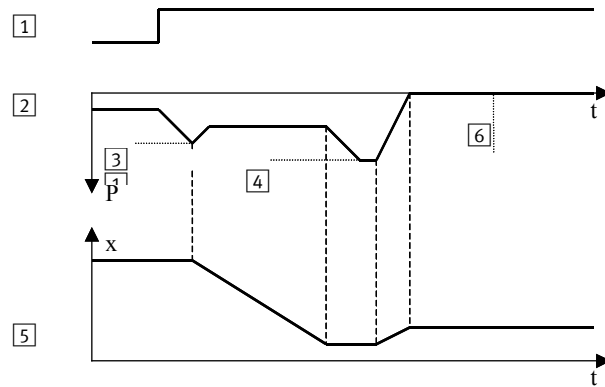


Fig. 9.18: Reference run

Welding

Standstill detection

During force build-up, the gun arms could move due to deflection of the gun. The standstill detection value determines the threshold below which the velocity must fall in order for the “Force reached” signal to be generated.

The following applies:

Standstill detection	Severity level
0	Detection switched off. “Force reached” is output as soon as the welding force is reached
1 ... 5	Detection active. The “Force reached” signal is output as soon as the welding force is reached and the speed of the main cylinder falls below the limit.

Table 9.21: Standstill detection

The permitted limit falls with increasing value. The “Force reached” signal is output later as the filter factor rises.

Teach distance	Function
0	The teach distance is calculated and specified by the controller. C guns 10 mm X gun at least 5 mm cylinder stroke Example: X gun translation ratio 1:3, cylinder stroke = 5 mm, teach distance = 15 mm
1	The entered teach distance is approached

Table 9.22: Teach distance

Weight force correction	Function
0	Function switched off
1	The friction and the arm weight is measured dependent on the space situation of the gun and compensated for each welding point.

Table 9.23: Weight force correction

Fast gun closing	Function
0	Function switched off
1	The closing process from the teach distance to the component is executed more quickly. The weight force correction function must be switched off for that purpose.

Table 9.24: Fast gun closing

7 axis

Entry of parameters for following error compensation in 7 axis operation

Parameter	Unit	Significance
Kv static	-	Constant of compensation switching dependent on the speed.
Ka static	-	Constant of compensation switching dependent on the acceleration.
Kv dynamic	-	Dynamic of compensation switching dependent on the following error.

Table 9.25: 7 axis

Fieldbus

Parameter	Unit	Significance
Swap bytes	-	Even and odd bytes are exchanged during transmission via fieldbus. To check, observe the life bit on the fieldbus of the robot.

Table 9.26: Fieldbus

Options

Parameter	Unit	Significance
Limit switch equalizer back	-	The initial position of the gun can also be sensed via a limit switch at input 2 of the controller. The fieldbus signal "Equalizer is back" is only set if input 2 is assigned and the actual pressure at the equalizer cylinder is less than -4.5 bar
Sensor check off	-	Deactivate monitoring of the position sensor.

Table 9.27: Options

9.9 Axis calibration

Axis calibration is used to determine the zero position and translation ratio between cylinder stroke and electrode stroke. The translation ratio for C-guns is 1.0. The translation ratio for X-guns is greater than 1.0, dependent on the arm length.



Incorrect processing of the axis calibration can lead to collisions and damage in the production plant. Always check the determined values for translation and maximum electrode opening.
The system is permanently under pressure during axis calibration. Never reach between the electrodes when measuring the electrode distance!



Axis calibration is to be carried out independently of the gun geometry (C/X). Before performance of the axis calibration, **new electrode caps** must be mounted. Care must be taken that the electrode shafts are in a good status

Terms	Significance
Zero position	The zero position is determined in axis calibration and saved in the controller. The zero position is the cylinder position with closed electrodes. If a setpoint position of 0 mm is specified, the electrodes of the gun are closed.
Translation ratio 1	Translation ratio 1 describes the ratio between the electrode position and cylinder position. The translation ratio is dependent on the measured electrode opening during axis calibration. The translation ratio is determined near the zero position, so the gun works precisely with small openings.
Translation ratio 2	The translation ratio 2 is the ratio between the maximum electrode opening and related cylinder stroke. The translation ratio 2 is calculated during axis calibration and determined by the operator by measuring the inside width at maximum gun opening.
Position values of the robot or WinSPZ	All position information that the robot sends over the fieldbus interface or is specified via WinSPZ refers to the electrode position. The closed gun has the position "0 mm".
Maximum electrode distance	The parameter should be considered a software limit switch. If this limit is exceeded, the error "Position not reachable" is reported.

Table 9.28: Axis calibration terms



Due to the mechanical conditions with X guns, the translation ratio changes dependent on the opening angle. The actual electrode position in X guns may differ from the measured opening if the opening angle is increasing.

The following conditions must be fulfilled prior to execution of axis calibration:

- PC control active,
- System error-free,
- Load voltage US2 on.

9.9.1 “Axis calibration” dialog box

The process of axis calibration is performed through this dialog box.

axis calibration

step 1: Set the cylinder stroke and start axis calibration

cylinder stroke after touching the electrodes

10.0 mm

step 2: measure the electrode distance

measured electrode distance

10.0 mm

step 3: maximum distance of the electrodes

maximum cylinder stroke

123.6 mm

step 4: measure the maximum electrode distance

maximum electrode distance

0.0 mm 124.0 mm

Fig. 9.19: “Axis calibration” dialog box

Input field	Unit	Significance
Cylinder stroke	mm	The cylinder opens by this stroke after determination of the zero position. Standard value 10 mm
Measured electrode stroke	mm	Input field for the electrode distance measured at the gun. Note: Translation ratio 1 = electrode stroke / cylinder stroke
Maximum cylinder stroke	mm	Maximum cylinder stroke between zero position and end position of cylinder. This value is determined automatically through Start step 3.
Maximum electrode stroke (display)	mm	Maximum electrode stroke calculated Maximum electrode stroke = Maximum cylinder stroke * translation ratio 1
Maximum electrode stroke (input)	mm	Input field for the measured maximum electrode opening of the gun. Note: Translation ratio 2 = Max. electrode opening / max. cylinder stroke

Table 9.29: Input fields in the “Axis calibration” dialog box

Button	Function
[Start] Step 1	The zero point is determined. The gun closes until the electrodes touch. Then the gun opens by the entered cylinder stroke.
[Start] Step 3	The gun is opened to a maximum to determine the maximum electrode opening.
[OK]	The values are transmitted to the controller and stored in the EEPROM.
[Cancel]	The values are discarded.

Table 9.30: Buttons in the “Axis calibration” dialog box



Note: Always carry out the axis calibration completely. The fieldbus output signal “Gun referenced” to the robot shows that the axis calibration has been performed completely.

9.9.2 Performance and process of axis calibration

Follow the process described in the Commissioning chapter under path calibration.



Warning

Danger of injury.

Attention: During the measurements, the main cylinder is under pressure. The pressure in the main cylinder is confined by the pneumatic auxiliary circuit in the cylinder. Do not remove any hoses or cables during the measurement process.



Carry out axis calibration with new electrode caps only.

Note:

Friction detection is reset during axis calibration – the friction factor is determined again.

9.10 Force calibration

Force calibration is used to scale the electrode force. To do this, the corresponding cylinder force is determined for two electrode forces. Two calibration points are enough for an exact force characteristics curve because the behaviour is linear between pressure and force. In force scaling, the current friction and weight force are also measured and stored. These values are used to compensate the weight force for each welding point. The robot sends the desired electrode force for each welding spot. The deflection of the gun is used for measuring the component thickness more accurately.

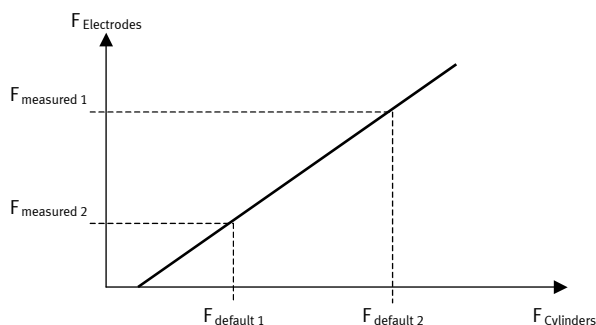


Fig. 9.20: Force calibration diagram

9.10.1 “Force calibration” dialog box

Force calibration is performed via this dialog box.

Fig. 9.21: “Force calibration” dialog box

Force calibration is used to scale the electrode force.



In force calibration, high forces can affect the gun and measuring instrument. The maximum electrode force of the gun must not be exceeded.



Recommended forces for calibration:

- Value 1 = at 1000 N electrode force
- Value 2 = at nominal force -10 % of the gun (see rating plate)

Input field / button	Unit	Significance
Preselected cylinder force	N	Specification of the cylinder force.
Measured electrode force	N	Input of the measured electrode force. Input of ON switches off the compensation of the weight force of the moving electrodes
Bending gun	mm	Field for bending of the gun arms during force build-up. These values are automatically entered after release of the button “Start force”. Note: To determine the bending, the dynamometer must be removed.

Table 9.31: Force calibration

Button	Function
[Value 1] / [Value 2]	Selection of calibration point.
[Open gun 60 mm]	Opens the electrodes to 60 mm
[Force build-up]	Closes the gun with slow speed and builds up the preset cylinder force.
[Open gun 20 mm]	Opens the electrodes to 20 mm.
[Weight]	The friction and weight force is determined and entered in the measured weight field.
[Ok]	Applies the force calibration data to the project. When the online mode is active, the calibration data determined is saved in the controller automatically.
[Cancel]	Cancels the force calibration data. The data is discarded; it is not applied to the project, and calibration data is not transmitted to the controller.

Table 9.32: Buttons in “Force calibration” dialog box

9.10.2 Performance and process of force calibration

Follow the process described in the Commissioning chapter under force calibration.

9.11 Trace

With WinSPZ, a trace can be recorded over 4 seconds.



Starts the trace at the next event in the control panel.



Starts the trace immediately for 4 seconds after the button in the toolbar is actuated.

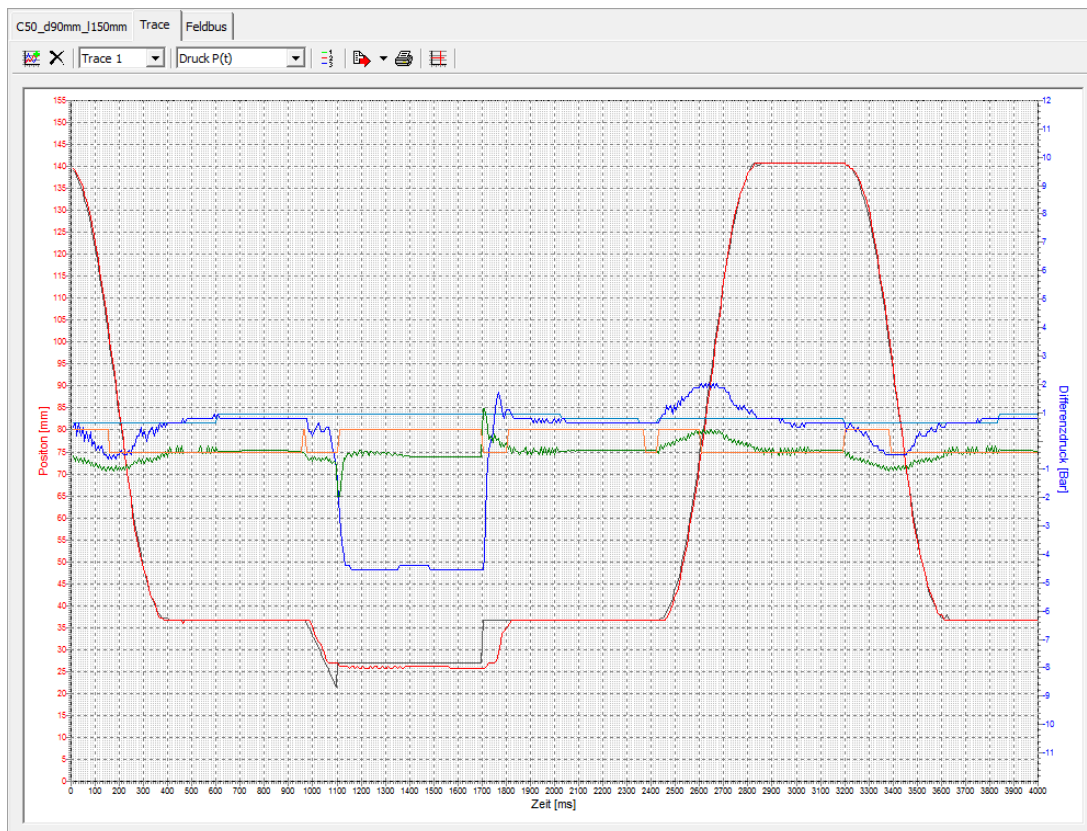


Fig. 9.22: Trace recording

Variable	Colour	Significance
Setpoint position cylinder	Black	Setpoint position of the main cylinder.
Actual position cylinder	Red	Actual position of the main cylinder.
Control signal main cylinder valve	Green	Mid-position is 75 mm max deflection +/- 10 mm grid.
Setpoint pressure / setpoint force	Blue	Setpoint value of differential pressure or cylinder force.
Actual pressure / actual force	Light blue	Actual value of differential pressure or cylinder force.
Position reached / Force reached	Orange	Signal is dependent on the selection of pressure or force.

Table 9.33: Displayed data in the trace

9.12 Fieldbus

In the fieldbus dialog, the data exchange between the higher-level robot and the controller is displayed.

Left column: data from the robot to the controller.

Right column: data from the controller to the robot.

C50_d90mm_1150mm	trace	field bus
0.00	● Error reset	● Gun Ready
0.01	● Energy saving on	● Energy saving is on
0.02	● Release regulator off	● Release regulator is off
0.03	● without force	● With force
0.04	● With TCP correction	● Handshake bit TCP OK
0.05	● Start learning teach position	●
0.06	● Take over teach position	● Teach position valid
0.07	● Robot manual	● Control manual
0.08	● Robot automatic	● Control automatic
0.09	● Reference	● Gun is referenced
0.10	● Release positioning	● Positioning active
0.11	● Set position valid	● Actual position valid
0.12	● Open gun	● Gun opened
0.13	● Close gun	● Force reached
0.14	● Gun jog open	● Equalizer is back
0.15	● Gun jog close	●
1.00	● Set value E-Force v1*100N	● Actual value E-Force v1*100N
1.01	● Set value E-Force v2*100N	● Actual value E-Force v2*100N
1.02	● Set value E-Force v4*100N	● Actual value E-Force v4*100N
1.03	● Set value E-Force v8*100N	● Actual value E-Force v8*100N
1.04	● Set value E-Force v16*100N	● Actual value E-Force v16*100N
1.05	● Set value E-Force v32*100N	● Actual value E-Force v32*100N
1.06	● Set value E-Force v64*100N	● Actual value E-Force v64*100N
1.07	● Set value E-Force v128*100N	● Actual value E-Force v128*100N
1.08	● Equalizer pressure v1	● Offset fixed E-Arm 1x0,1mm
1.09	● Equalizer pressure v2	● Offset fixed E-Arm 2x0,1mm
1.10	● Equalizer pressure v4	● Offset fixed E-Arm 4x0,1mm
1.11	● Equalizer pressure v8	● Offset fixed E-Arm 8x0,1mm
1.12	● Equalizer pressure v16	● Offset fixed E-Arm 16x0,1mm
1.13	● Equalizer pressure v32	● Offset fixed E-Arm 32x0,1mm
1.14	● Equalizer pressure v64	● Offset fixed E-Arm 64x0,1mm
1.15	● Equalizer pressure v128	● Offset fixed E-Arm 128x0,1mm
2.00	● Set position v 1x0,1mm	● Actual position v 1x0,1mm
2.01	● Set position v 2x0,1mm	● Actual position v 2x0,1mm
2.02	● Set position v 4x0,1mm	● Actual position v 4x0,1mm
2.03	● Set position v 8x0,1mm	● Actual position v 8x0,1mm
2.04	● Set position v 16x0,1mm	● Actual position v 16x0,1mm
2.05	● Set position v 32x0,1mm	● Actual position v 32x0,1mm
2.06	● Set position v 64x0,1mm	● Actual position v 64x0,1mm
2.07	● Set position v 128x0,1mm	● Actual position v 128x0,1mm
2.08	● Set position v 256x0,1mm	● Actual position v 256x0,1mm
2.09	● Set position v 512x0,1mm	● Actual position v 512x0,1mm
2.10	● Set position v 1024x0,1mm	● Actual position v 1024x0,1mm
2.11	● Set position v 2048x0,1mm	● Actual position v 2048x0,1mm

Fig. 9.23: Status of fieldbus signals

9.13 Control of the welding gun

All functions of the gun can be tested and simulated with the help of the WinSPZ software.

9.13.1 Functions of the Control panel “Experts”

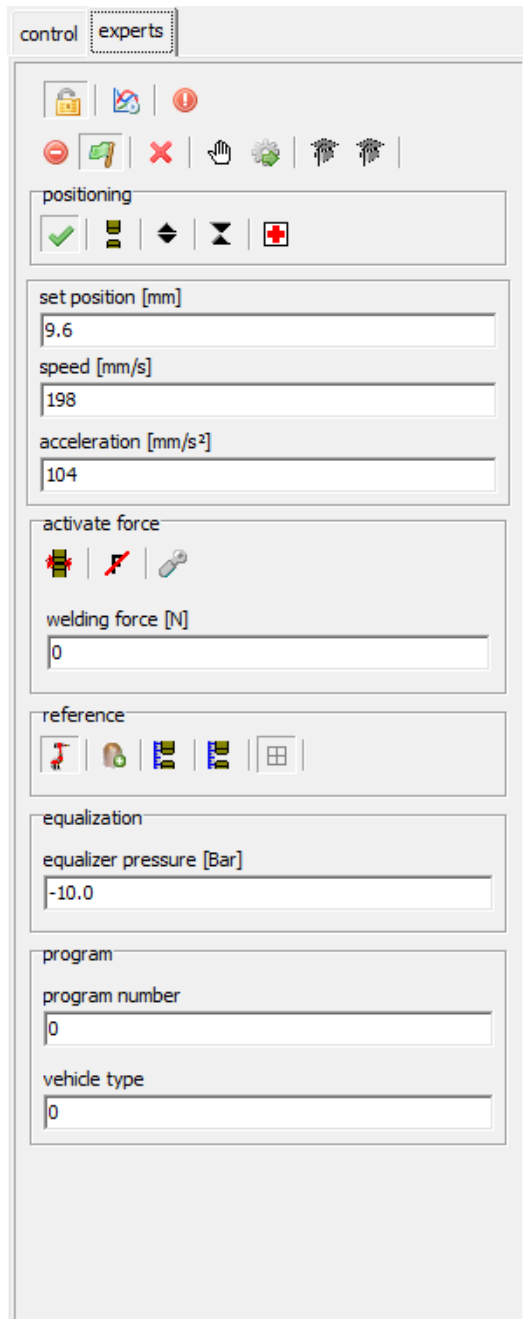


Fig. 9.24: Control panel “Experts” dialog

PC control



A mouse click on this button switches control of the drive to the PC. Control by the PC is indicated by the pressed button. Clicking the pressed button again switches control of the drive back to the fieldbus master.



Warning

Sudden uncontrolled movements of the welding gun can cause injuries. When control of the drive is switched over to the fieldbus, the values set through the fieldbus apply immediately (e.g. position values).



If “PC control” is still active when the software package WinSPZ is exited, PC control is disconnected automatically after 1 minute and is switched back again to the signals of the fieldbus. This can cause movements of the gun.

9.13.2 Description of the control bits of the robot

No.	Bit	Description
1	PC control 	If the PC control is activated and the load voltage US2 is on, the gun can be controlled through the WinSPZ software and placed in operation. Additional input fields and control buttons are switched to active. If the connection is interrupted, the PC controller switches off automatically after 1 minute. -> See comments on page 29
2	Error reset 	Acknowledge the currently present error message.
3	Robot manual 	The robot is in manual override. Positioning with “gun open” is carried out at reduced speed.
4	Robot automatic 	The robot is in automatic operating mode. Positioning with “gun open” is carried out at maximum speed.
5	Start learning teach position 	The robot starts the teach run to the component and opens to the pre-defined cylinder stroke. C-gun: 10 mm electrode stroke X-gun: 5-10 mm cylinder stroke, dependent on the translation ratio results in an electrode stroke ≥ 10 mm
6	Teach position is accepted 	The robot shows that the teach position has been accepted.
7	Release positioning 	If this bit is set, the gun has movement release and reacts to the control signals of the robot, such as – Learn teach position – Close gun with force build-up – Open gun to setpoint position – Open / close gun pushing – Measure after milling – Measure before milling If this bit is off, the gun is in 7 axis operation and follows a valid setpoint position from the robot.
8	Setpoint position valid 	This bit indicates that the setpoint position of the robot is valid and the electrode may move to the preset position. If the “setpoint position valid” bit is on and the “release positioning” bit is off, the gun follows the position of the robot in 7 axis operation.
9	Close gun pushing 	The gun closes with slow speed up to the mechanical limitation of the electrode.
10	Open gun pushing 	The gun opens with slow speed up to the mechanical limitation of the gun.
11	Open gun 	With the “gun open” control bit, the gun jumps to the preset setpoint position of the robot. The status signal “gun opened” indicates that the setpoint position has been reached.
12	Close gun 	With the “close gun” bit, the electrode is moved to the component. The preset electrode force is built up. The “force reached” bit signals the robot that the electrode force has been reached and the welding can be started.

No.	Bit	Description
13	Without force 	With this bit, the electrode is not closed at the signal “close gun”. Only the equalising function is activated.
14	Function test without component 	With this operating mode, the gun builds up a reduced force (electrode force function test). This function prevents the electrodes from sparking over during the function test of the gun without component.
15	Compensation by robot 	This bit is set standard by the robot to 1. The bit shows that the robot compensates for the cap wear of the fixed and movable electrode arm.
16	Caps changed 	This bit shows that the robot is driven to the wait position for cap dressing.
17	Measure before milling 	The caps are measured with this bit. The position of the electrode caps is compared to the zero position. If the deviation exceeds the tolerance of the geometry test parameter, the “error cap dressing geometry” is displayed. The status signal “cap dressing OK” displays to the robot that the change was successful. The cap abrasion of the fixed E arm is set to zero.
18	Measure after milling 	With this bit the maximum abrasion of the caps is checked after each milling cycle. If the minimum wear is reached, the upcoming cap replacement is signalled to the robot with the cap wear error message.
19	With TCP 	If this bit is set, the abrasion of the fixed electrode cap is sent to the robot through the fieldbus.
20	Controller enable off 	With this bit, controller enable is switched off. With the status bit “controller enable is off”, the robot is signalled that the gun can now be undocked.
21	Diagnostic function 	With this bit and the “release positioning” signal, a function test for the shut-off valves of the main cylinder is started with the diagnostic function.

Table 9.34: Control bits from the robot

9.13.3 Data from the robot

Input field	Unit	Description
Setpoint position	mm	The setpoint position describes the electrode position of the gun. At position 0 mm (zero position), the gun is closed. When you confirm your entry with the RETURN key, this setpoint is sent to the controller. 1 bit / 0.1 mm
Welding force	N	Electrode force for force build-up during “close gun”. 1 bit = 100 N
Equalizer pressure	bar	Setting of the differential pressure in the equalising cylinder during welding. Note: The binary specification value for the robot can be read out in the window [Setpoint values].
Program number		Program number currently selected by the robot. Based on this program number, the switching points stored in the controller are selected during movement. (Switching point: electrode position for switching from fast speed to slow speed when the gun is closing) Note: For the “close gun” function to be executed in automatic mode, the robot must choose a program number and a vehicle type. Service programs: - Program number: 1 -98 - Vehicle type: 0 Welding programs: - Program number: 99 -255 - Vehicle type: 1 -255
Vehicle type		See program number

Table 9.35: Data from the robot

9.13.4 Data from WinSPZ

Input field	Unit	Description
Speed (only WinSPZ)	mm/s	Cylinder speed for test of the 7 axis operation via WinSPZ with PC control. With these variables, the speed of the internal ramp generator can be specified.
Acceleration (only WinSPZ)	mm/s ²	Cylinder acceleration for test of the 7 axis operation via WinSPZ with PC control. With these variables, the acceleration of the internal ramp generator can be specified.

Table 9.36: Data from WinSPZ

9.13.5 Functions of the control panel

The screenshot shows the 'control' tab of the WinSPZ software interface. It features a toolbar with icons for enabling trace, acknowledging error, and selecting operation mode. The 'positioning' section allows switching between 'PtP' (Point-to-Point) and 'A7' (Axis 7) modes, with corresponding jog and position setting buttons. The 'force build up' section includes fields for two different force levels. The 'equalization' section has a single pressure setting field. The 'test run' section includes fields for time interval, set cycles, and actual cycles, along with start and stop buttons.

Fig. 9.25: Control dialog

Function	Description
	Enable trace function. The trace is automatically started at the next action.
	Acknowledge error
Point-to-point Point-to-point operation	The gun moves at maximum speed to position 1 or position 2.
A7 Axis 7	The gun positions itself with the preset speed and acceleration through a ramp generator to position 1 or position 2.
Pushing open Pushing closed	Open / close gun pushing.
Position 1 [mm] Position 2 [mm]	The specified position is travelled to through actuation of the button.
Force 1 [N] Force 2 [N]	The specified force is built up through actuation of the button.
Equalization [bar]	Through actuation of the button, the specified equalization pressure is output.

Function	Description
Start of continuous operation	Starts the continuous operation cycle for testing the gun. The selected fields (position 1/2 force 1/2) are called up after the interval time [ms]. Continuous operation stops when the actual cycles have reached the preset setpoint cycles.
Continuous operation stop	Stops the continuous operation

Table 9.37: Control dialog

9.13.6 Status bits to the robot

No.	Bit	Description	Condition
1	Ready	Displays that the gun is ready for positioning.	Protection circuit closed, gun referenced, no error, controller enable on.
2	Gun is referenced	Shows that the axis calibration has been performed.	Becomes zero if the axis calibration is not performed completely or the translation ratio was changed through input of new parameters. Perform axis calibration again.
3	Equalizer back	Gun body in initial position.	Equalizer pressure actual value < -4.5 bar. A limit switch can be connected at input 2. See WinSPZ.
4	Manual control	Control in PC operation.	WinSPZ PC control active.
5	Automatic control	Controller reacts to commands from the fieldbus.	WinSPZ PC control off.
6	Force reached	Setpoint force reached during close gun.	Setpoint force – actual force < 100 N
7	Teach position valid	Indicates that the valid teach position is present at the actual position status word.	Component sensed and gun opened to teach dimension.
8	Controller enable is off	“Controller enable off” controller signal is requested. Gun can be undocked.	Controller disabled.
9	Handshake TCP OK	Shows that a valid value is present for the TCP offset for the fixed E-arm.	Status bit “with TCP” from the robot on.
10	Positioning active	Indicates that positioning release is present.	“Positioning release” status bit on.
11	Actual position valid	Always 1	Always 1
12	Life bit gun controller	Shows the robot that the controller and the fieldbus are running.	Flashes at 1-second pulse.
13	Cap change OK	Shows that the function “Measure before milling” has been carried out successfully.	Tolerance for geometry check OK.
14	Milling OK	Indicates that the function “Measure after milling” has been carried out successfully.	Tolerance after milling OK.
15	Measuring done	Indicates that the function “Measure before milling” or “Measure after milling” has been ended.	Process ended.
16	Invalid gun code	Shows that the gun code specification from the robot is invalid.	The code stored in the controller does not agree.

Table 9.38: Status bits for the robot

9.13.7 Data for the robot

Input field	Unit	Description
Current position	mm	Current electrode opening. 1 bit / 0.1 mm
Actual force E-force	N	Current electrode force with “Close gun”. 1 bit = 100 N
Reset value TCP	mm	Display of cap wear of the electrode cap of the fixed electrode arm.
Device code		Festo always reports code 6.

Table 9.39: Data for the robot

9.14 Functions of the gun

9.14.1 Switching points

The switching points are selected by the robot through specification of the program number. The gun travels with maximum speed to the switching point and then reduces its speed.

Program number	Switching point	Program in the robot
1-10	40 mm	Force scaling
11-20	20 mm	Current scaling
21-30	10 mm	Cap setting after cap dressing
31-40	20 mm	Cap milling
41-50	5 mm	Short circuit welding after start milling
51-60	5 mm	Short circuit welding after normal milling
Other	0 mm	

Table 9.40: Switching points

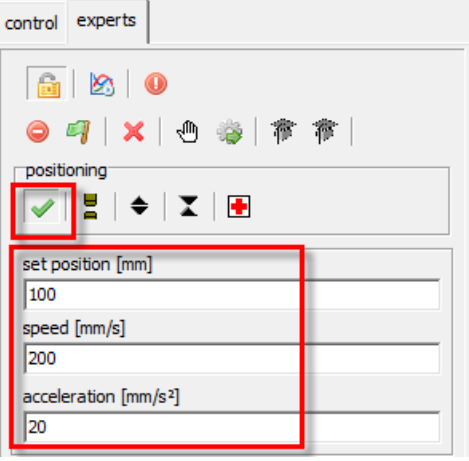
9.14.2 Teach welding point

The operator teaches each welding point. The component is scanned and the gun opens to the Teach distance. This distance is 10 mm for C gun and greater than or equal to 10 mm for X guns. But the electrode opening must be at least 5 mm of cylinder stroke. This position is stored in the robot for the respective welding point. The robot positions to this distance from the welding point.

No.	Robot	Gun
1	Operator is positioning on welding point. Release positioning = 1 Start teach position learning = 1 Test via WinSPZ. 	
2		Enable equalizer. Open electrode position to 25 mm. Close with constant speed. Sense component. Open = 10 mm electrode stroke (For X-guns at least 5 mm of cylinder stroke) Equalizer back. Teach position valid = 1 Output teach position as actual position. Wait for signal "Take over teach position".
3	Read actual position. Teach position is accepted = 1 Test via WinSPZ. 	
4		Teach position valid = 0 Output actual position.

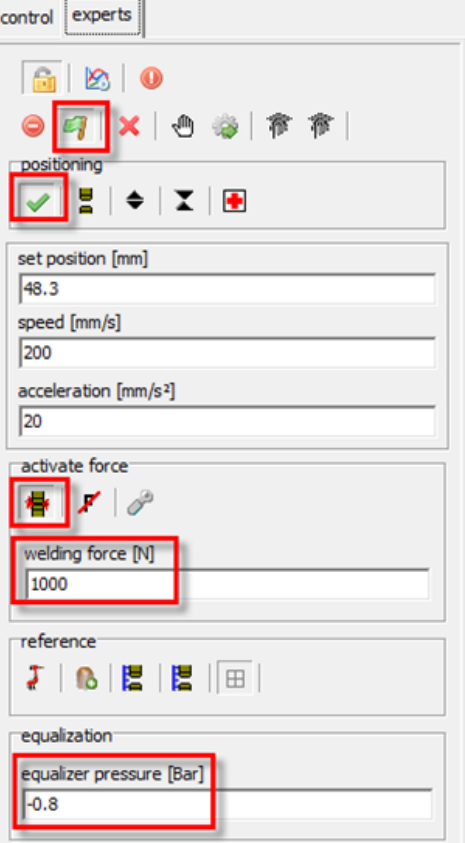
9.14.3 7 axis operation

The robot validly displays the 7 axis operation to the gun with the “setpoint position” bit. The gun follows the position of the robot as an external axis. The robot moves to points in the space via a ramp through the path planner. The robot thereby accelerates to a maximum speed. The gun must follow the robot with minimum deviation. The maximum following error is monitored in the robot. Through a service program, the robot determines the dead time of the position control. It measures how many interpolation cycles are required until the robot gets the specified setpoint position reported back as an actual position. The gun follows the robot only if the “Gun referenced” signal and the “Ready” signal are reported to the robot.

No.	Robot	Gun
1	Release positioning =0 Setpoint position valid = 1 Setpoint position as trajectory. Robot accelerates to a maximum speed. Robot monitors following error. Test via WinSPZ.	
2		The “Gun ready” and “Gun referenced” signal must be set. The gun follows the setpoint position of the robot. The following error is minimised.

9.14.4 Close gun

With the signal “Close gun”, the gun is closed and the setpoint force is built up. The gun reports “Force reached” to the robot. The welding controller sets the welding point. The weight force is measured and compensated for each welding point. The equalizer is activated by the robot with the binary value. The gun can be closed from any desired position.

No.	Robot	Gun
1	Specify equalization pressure. Specify setpoint force. Release positioning =1 Close gun = 1 Check program number / vehicle type. Test with WinSPZ. 	
2		Check setpoint force limit. Enable equalizer. Fast positioning to switching point. Close with constant speed. Measure friction and weight force. Detect touching point on component. Correct weight force. Switch to force control. Gun opened = 0 Force reached = 1 Equalizer is back = 0
3	Start welding	

9.14.5 Open gun

After welding is completed, the robot sets the signal “gun open”. The gun positions itself to the current set position and, after reaching the set position, sets the signal “Gun opened”.

No.	Robot	Gun
1	Specify set position. Release positioning = 1 Position valid = 1 Open gun = 1 Test with WinSPZ. 	
2		Check set position limit. Approach set position with max. speed. Equalizer in initial position. Position reached? Gun opened = 1 Force reached = 0 Equalizer is back = 1
3	Robot positioned on the next welding point	

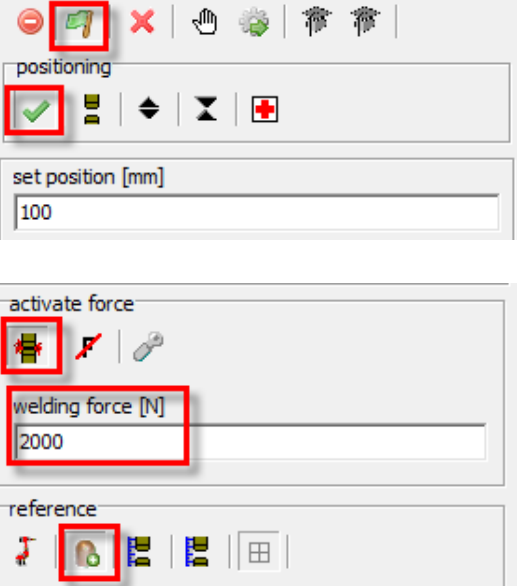
9.14.6 Open / close gun pushing

With the function “Open pushing” and “Close pushing”, the gun can be travelled slowly open and closed. In addition, the gun can be travelled to the end position. If the actual position is changed, the robot must then synchronise to the new actual position in order to start synchronously in the 7 axis operation.

No.	Robot	Gun
1	Release positioning =1 Close gun pushing = 1 Test via WinSPZ. 	
2		Gun closes as long as button is actuated.
	Release positioning =1 Open gun pushing = 1 Test via WinSPZ. 	
		Gun opens as long as button is actuated.

9.14.7 Set caps

The function “Set caps” serves to press the electrode caps on after cap dressing. The position of the caps is compared to the zero position. If the parameter “Tolerance set caps” is exceeded, the “Geometry error” error message follows.

No.	Robot	Gun
1	Specify setpoint force. Release positioning = 1 Close gun = 1 Caps changed = 1 Test via WinSPZ. 	
2		Check setpoint force limit. Fast positioning to switching point. Close with constant speed. Detect touching point on component. Check “Tolerance set caps”. Cap dressing OK= 1 // if geometry OK Switch to force control. Gun opened = 0 Force reached = 1

9.14.8 Measure before milling / geometry check

The geometry check is performed after the “Set caps” function of the robot. The caps are measured and the tolerance is checked against the zero position. The gun reports “Cap change OK”. A check is made whether the caps were replaced after maintenance and that the maximum shaft lowering has not been exceeded. The reset value for the robot is set to zero.

No.	Robot	Gun
1	Release positioning =1 Caps changed = 1 Measure before milling = 1 Test via WinSPZ. 	
2		Fast positioning to switching point. Measure electrode caps. End of measuring = 1 Gun opened = 1 Cap dressing OK= 1 // if tolerance geometry check OK. Save position of new caps. Reset value TCP = 0 Open to set position.

9.14.9 Measure after Milling

Start milling

When measuring after start milling, the caps are measured and examined whether the tolerance during milling was maintained. If the measurement is within tolerance, the gun reports “Milling OK”.

No.	Robot	Gun
1	Release positioning = 1 Caps changed = 1 Measure after milling = 1 Test via WinSPZ 	
2		Fast positioning to switching point Measure electrode caps End of measuring = 1 Save position of new caps Milling OK = 1 // if milling tolerance OK Calculate reset value TCP Open to set position

Normal milling

After each normal milling, the caps are measured. If the measurement is within tolerance, the gun reports “Milling OK”. If the life of the caps is reached, the gun reports cap wear.

No.	Robot	Gun
1	Release positioning = 1 Caps changed = 0 Measure after milling = 1 Test via WinSPZ 	
2		Fast positioning to switching point Measure electrode caps End of measuring = 1 Milling OK = 1 // if milling tolerance OK Error cap wear = 1 // if cap life is reached Calculate reset value TCP Open to set position

9.14.10 Diagnostic function for checking the blocking function in the main cylinder

The gun controller performs a functional test of the blocking valves 1V2, 1V3, 1V4. This functional test is controlled by the robot by setting the bit “diagnostic function”. The blocking function is activated. The proportional directional control valve 1V1 is actuated. If no pressure and position change takes place in the main cylinder 1A1, the blocking function is working properly.

Note: The bit “Diagnostic function error” remains stored in case of loss of voltage. The bit “Diagnostic function error” is reset if the diagnostic function has been run through successfully.

No.	Robot	Gun
1	Release positioning = 1 Diagnostic function = 1 Test via WinSPZ. 	Diagnostic function IO = 0 Diagnostic function error = 0
2		Blocking valves 1V2, 1V3, 1V4 are switched. The connections 2 and 4 of the main cylinder 1A1 are pressurised. The pressure and position change is checked in the main cylinder 1A1. <u>Diagnostic I/O:</u> Diagnostic function I/O = 1 Common error = 0 Gun ready = 1 <u>Diagnostic error:</u> Diagnostic function error = 1 Common error = 1 Gun ready = 1 Error is saved in EEPROM. Error can only be acknowledged through successful run through the diagnostic function.

9.15 Set / actual values panel

The “Set /actual values” panel displays the cyclical data in online mode.

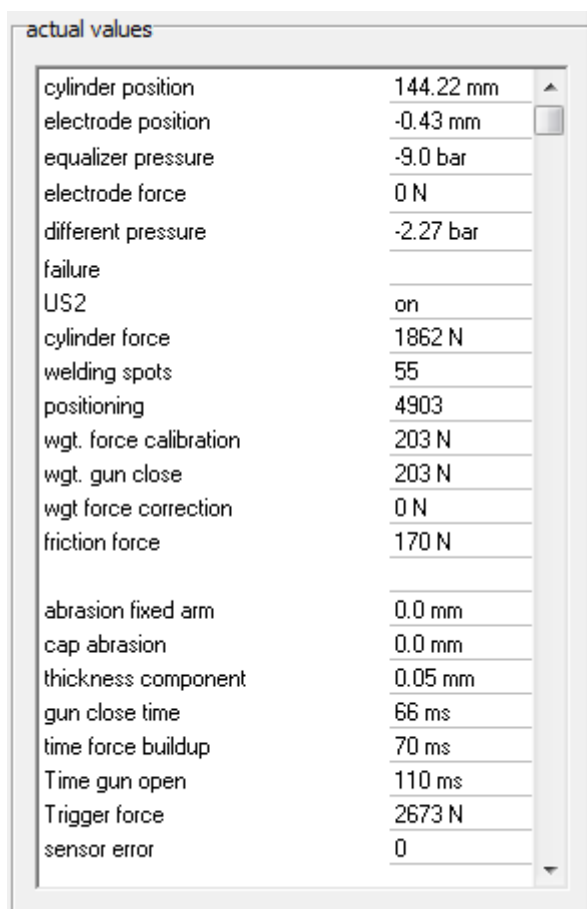


Fig. 9.26: Set / actual values panel

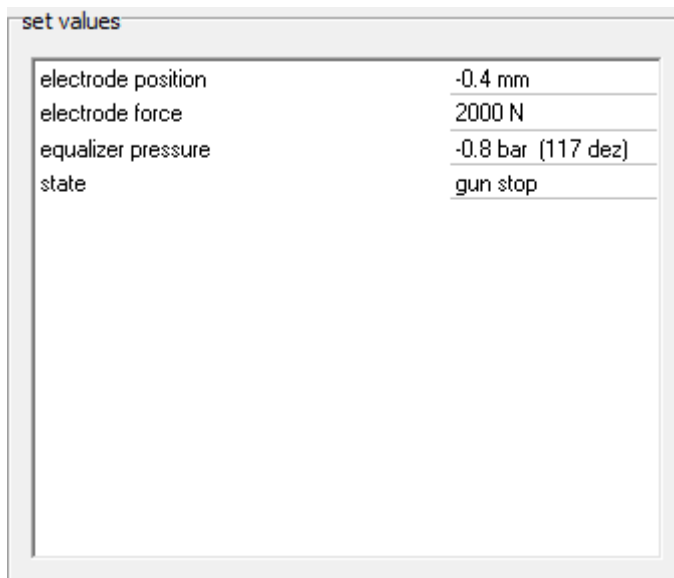
9.15.1 Actual values

No.	Value	Comment
1	Cylinder position	Actual position of the cylinder
2	Electrode position	Current electrode opening
3	Equalizer pressure	Current differential pressure in the equalizer cylinder. If the gun is in the initial position, the currently applied operating pressure is displayed (but maximum 10 bar).
4	Electrode force	Current electrode force. This is sent to the robot with a resolution of 1 bit / 100 N.
5	Different pressure	Differential pressure in the main cylinder.
6	Failure	Current error message in clear text.
7	US2	Status of load voltage (protective circuit). The gun may perform movements only if the US2 is displayed as on.
8	Welding spots	Number of welding points.
9	Positioning	Number of positionings.
10	Wgt. force calibration	Measured weight and friction force during force calibration.
11	Wgt. gun close	Measured weight and friction force during closing of the gun.
12	Wgt force correction	Corrected cylinder force with “Close gun”.
13	Friction force	Measured friction force during “Close gun”.
14	Abrasion fixed arm	Cap wear of the fixed gun arm.
15	Cap abrasion	Total cap wear.
16	Thickness component	Measured thickness of component.

17	Gun close time	Time from start of “Close gun” until fitted onto the component.
18	Time force build-up	Time from fitting onto the component until force build-up and output of “Force reached” signal
19	Time gun open	Time from the signal “Open gun” until “Gun is opened” is output
20	Sensor error	Displays how often the position sensor generated a measurement error.

Table 9.41: Actual values

9.15.2 Set values

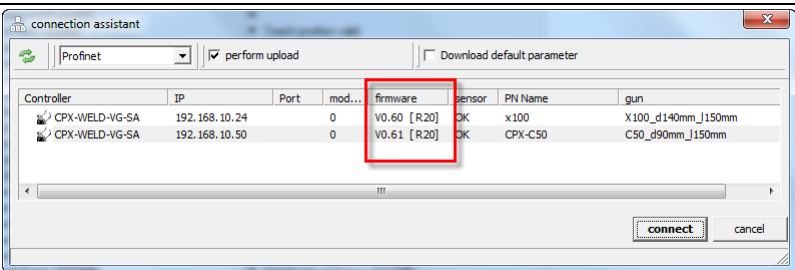
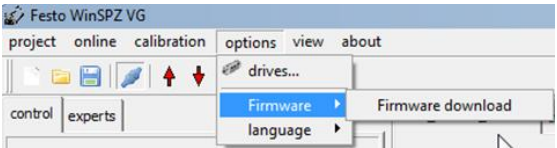
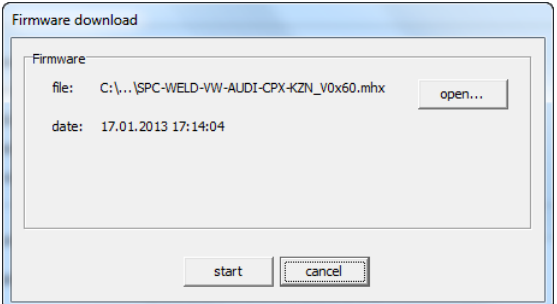
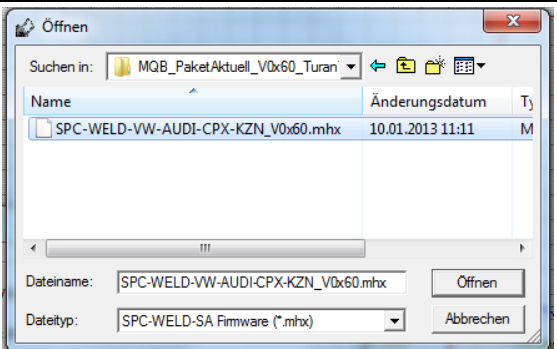
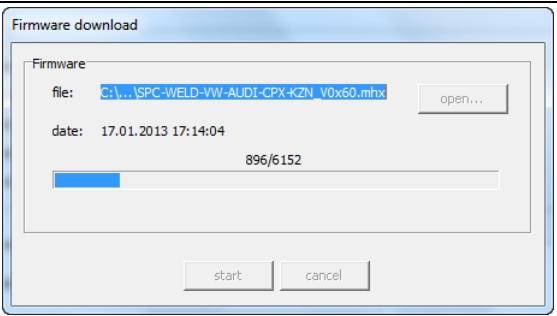


No.	Value	Comment
1	Electrode position	Setpoint position of the robot.
2	Electrode force	Setpoint force of the robot.
3	Equalizer pressure	Equalizer pressure from the robot.
4	State	Current selection of the operating mode of the robot.

Table 9.42: Set values

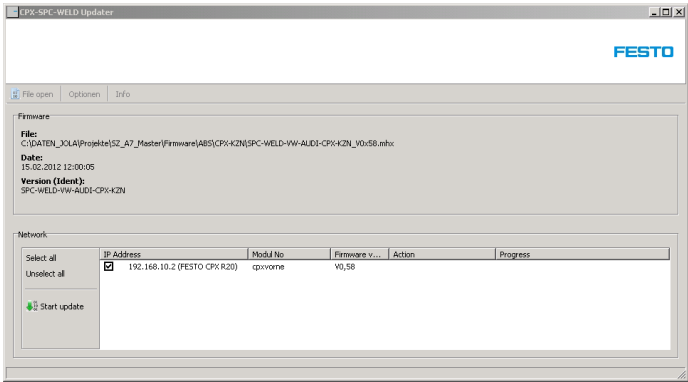
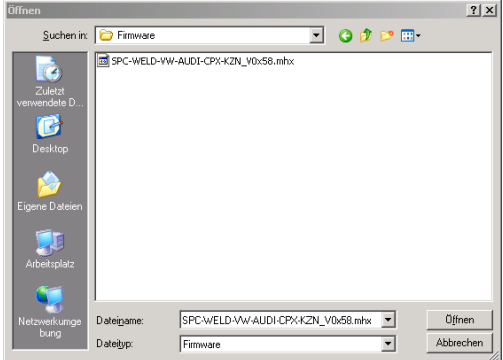
9.16 Firmware update

This firmware can be loaded to the controller by using WinSPZ.

No.	Action	Comment																								
1	Establish a connection to the controller.																									
2	The installed firmware version is displayed.	 <table><tr><th>Controller</th><th>IP</th><th>Port</th><th>mod...</th><th>firmware</th><th>sensor</th><th>PN Name</th><th>gun</th></tr><tr><td>CPX-WELD-VG-SA</td><td>192.168.10.24</td><td>0</td><td></td><td>V0.60 [R20]</td><td>OK</td><td>x100</td><td>X100_d140mm_1150mm</td></tr><tr><td>CPX-WELD-VG-SA</td><td>192.168.10.50</td><td>0</td><td></td><td>V0.61 [R20]</td><td>OK</td><td>CPX-C50</td><td>C50_d90mm_1150mm</td></tr></table>	Controller	IP	Port	mod...	firmware	sensor	PN Name	gun	CPX-WELD-VG-SA	192.168.10.24	0		V0.60 [R20]	OK	x100	X100_d140mm_1150mm	CPX-WELD-VG-SA	192.168.10.50	0		V0.61 [R20]	OK	CPX-C50	C50_d90mm_1150mm
Controller	IP	Port	mod...	firmware	sensor	PN Name	gun																			
CPX-WELD-VG-SA	192.168.10.24	0		V0.60 [R20]	OK	x100	X100_d140mm_1150mm																			
CPX-WELD-VG-SA	192.168.10.50	0		V0.61 [R20]	OK	CPX-C50	C50_d90mm_1150mm																			
3	Open menu Options -> "Firmware download".																									
4	In the "Firmware download" dialog, actuate the "Open" button.																									
5	Select firmware file "xxx.mhx".																									
6	Actuate "Start"																									

9.17 Firmware update for multiple systems

The firmware can be loaded with the software “CPX SPC-Weld Updater” simultaneously for several systems over Ethernet.

No.	Action	Comment						
1	Start software “CPX SPC-Weld Updater”.	 <table border="1"> <thead> <tr> <th>IP Address</th> <th>Modul No</th> <th>Firmware v...</th> </tr> </thead> <tbody> <tr> <td>☒ 192.168.10.2 (FESTO CPX R20)</td> <td>cpxvorne</td> <td>V0,58</td> </tr> </tbody> </table>	IP Address	Modul No	Firmware v...	☒ 192.168.10.2 (FESTO CPX R20)	cpxvorne	V0,58
IP Address	Modul No	Firmware v...						
☒ 192.168.10.2 (FESTO CPX R20)	cpxvorne	V0,58						
2	All welding gun systems found on the network are displayed.	<table border="1"> <thead> <tr> <th>IP Address</th> <th>Modul No</th> <th>Firmware v...</th> </tr> </thead> <tbody> <tr> <td>☒ 192.168.10.2 (FESTO CPX R20)</td> <td>cpxvorne</td> <td>V0,58</td> </tr> </tbody> </table> Example shows: Firmware Servobox :V0.58 Firmware Fieldbus Node: Rev 20	IP Address	Modul No	Firmware v...	☒ 192.168.10.2 (FESTO CPX R20)	cpxvorne	V0,58
IP Address	Modul No	Firmware v...						
☒ 192.168.10.2 (FESTO CPX R20)	cpxvorne	V0,58						
3	Open file.	Select firmware file “xxx.mhx”. 						
4	Start update.	 Firmware is loaded.						

10 Diagnostics and error messages

10.1 Error messages

The following error messages are recorded in the controller. These error messages are reported over 16 bits to the robot. All messages are displayed in clear text on the WinSPZ. The error can be cleared through "Error reset". For some errors, the "Ready" signal to the robot is deleted.

Error number	Error text	Ready signal	Description	Error elimination
1	Measuring system error	0	Position signal outside the valid measuring range.	Check drive cable.
2	Reserve	1		
3	No pressure supply	0	If actual pressure in the main cylinder is less than 1 bar for 4 seconds. The error is automatically cleared if the pressure in both chambers of the main cylinder > 3 bar.	Check compressed air supply. Check tubing connection. Check function of the shut-off valves 1V3 and 1V4 in the main cylinder. Check output 0 at the controller and shut-off valve 1V2.
4	Maximum force exceeded	0	The maximum electrode force is exceeded during positioning of the electrodes. Actual force > Maximum electrode force	Check maximum electrode force parameter. Check for obstacle in the travel range of the electrodes.
5	Internal error	0		Servicing required
6	Position not reachable	1	Specify setpoint position outside the maximum opening dimension.	Specify a valid position value. Check axis calibration and maximum opening dimension parameter.
7	Pressure sensor in main cylinder is faulty	0	Actual pressure value outside the valid measuring range.	Check cable to the main cylinder. Servicing required / replace main cylinder.
8	Incorrect force specification	1	Force setpoint specification outside min. - max. electrode force.	Specify a valid force value. Check minimum / maximum electrode force parameter.
9	Positioning timeout	0	Setpoint position is not reached after 10 s.	Check for obstacle in the travel range. Check valve cable. Check function of the shut-off valves 1V3 and 1V4 in the main cylinder. Check output 0 at the controller and shut-off valve 1V2.
10	Force timeout	0	If the setpoint force is not reached after 2 seconds with "Close gun".	Check the supply pressure Check cylinder parameter Check MPYE valve Check force calibration

Error number	Error text	Ready signal	Description	Error elimination
11	Diagnostic function	1	If a pressure change or position change is detected with switched shut-off valves 1V2,1V3,1V4 in the main cylinder.	Check function of the shut-off valves in the main cylinder. Replace main cylinder, if necessary. Restart diagnostic function in order to acknowledge error.
12	Timeout equalizer	1	Equalizer pressure was not reached within 2 seconds.	Check cable to the MPYD. Check supply pressure at least 5 bar.
13	Parameter error	0	Invalid values for position controller.	Reload parameters via WinSPZ.
14	Max. friction exceeded	1	Friction force exceeded during "Close gun".	Determine reason for increased friction / re-lubricate if needed. Check force calibration. Check friction force error parameter.
15	Reserve	1		
16	Incorrect program number	1	Robot sends program number = 0 with Close gun signal.	Check specification of the program number by the robot.
17	Over-temperature transformer	1	Over-temperature transformer is checked and reported only for "Close gun".	Check input 0. Over-temperature transformer must be on.
18	Reserve	1		
19	Reserve	1		
20	Short circuit at output 0	0	Overload at the output	Eliminate short circuit
21	Short circuit at output 1	0	Overload at the output	Eliminate short circuit
22	Short circuit at output 2	0	Overload at the output	Eliminate short circuit
23	Short circuit at output 3	0	Overload at the output	Eliminate short circuit
24	Reserve	1		
25	Reserve	1		
26	Parameterisation error controller	0	Controller parameter $K_0 = 0$ or $K_p = 0$	Parameterise controller
27	Reserve	1		
28	Cap dressing	1	Tolerance dimension exceeded for geometry check with Set cap or Measure after milling.	Maximum shaft lowering exceeded Incorrect caps used Check parameter tolerance geometry check Check parameter tolerance set caps.
29	Cap abrasion	1	Maximum life of the caps is reached.	Wear dimension of the caps exceeded. Replace caps.

Error number	Error text	Ready signal	Description	Error elimination
30	Milling not OK	1	Tolerance during milling is checked after each measure after milling.	Chip adheres to cap. Check parameter maximum increase per milling cycle. Check parameter maximum reduction per milling cycle. Maximum milling of the caps exceeded or fallen below.
31	Cap loss	1	Cap lost during “close gun”.	Check whether the cap is missing.
32	Reserve	1		

Table 10.1: Error messages

10.1.1 Assignment of error bits on fieldbus

Error number	Error text	Ready signal	Preliminary warning gun	Warning, lubrication required	Force build-up too long	Program error / incorrect force	Target position invalid	Component failure	Gun moves hard	Milling not OK	Error cap wear	Error caps or geometry	Following error/max. force exceeded	Error equalization system	Error control/regulator	Error temperature transformer	Error compressed air supply	Common malfunction
1	Measurement system error	0													1			1
2	Reserve	1																
3	No pressure supply	0															1	1
4	Maximum force exceeded	0											1					1
5	Internal error	0													1			1
6	Position not reachable	1					1											1
7	Pressure sensor in main cylinder faulty	0													1			1
8	Incorrect force specification	1				1												1
9	Timeout positioning	0											1					1
10	Force timeout	1			1													1
11	Diagnostics function	1																1
12	Timeout equalizer	0												1				1
13	Parameter error	0													1			1
14	Friction force malfunction exceeded	1							1									1
15	Reserve	1											1					
16	Incorrect program number	1				1												1

Error number	Error text	Ready signal	Preliminary warning gun	Warning, lubrication required	Force build-up too long	Program error / incorrect force	Target position invalid	Component failure	Gun moves hard	Milling not OK	Error cap wear	Error caps or geometry	Following error/max. force exceeded	Error equalization system	Error control/regulator	Error temperature transformer	Error compressed air supply	Common malfunction
17	Overtemperature transformer	1														1		1
18	Reserve	1																
19	Reserve	1																
20	Short circuit at output 0	0													1			1
21	Short circuit at output 1	0													1			1
22	Short circuit at output 2	0													1			1
23	Short circuit at output 3	0													1			1
24	Reserve	1																
25	Reserve	1																
26	Parameterisation error controller	0													1			1
27	Reserve	1																
28	Cap dressing	1										1						1
29	Cap wear	1									1							1
30	Milling not OK	1								1								1
31	Cap loss	1										1						1
32	Reserve	1																
-	Warning friction limit exceeded	1	1	1														

Table 10.2: Assignment of error bits

10.2 Fault clearance on the system

No display on the controller

Cause	Remedy
Supply voltage is not correct or controller is defective	Check supply voltage

Main cylinder does not move

Cause	Remedy
Valve cable not connected	Check the connection valve – MPYE 1V1
Cable to the main cylinder not connected	Check the connection drive – main cylinder 1A1
Controller reports error	Check controller error messages
Gun reports no ready status signal	Check ready status
Gun reports no gun referenced status signal	Perform axis calibration
No compressed air supply	Check compressed air supply
Controller parameters are not set	Check parameterisation
US2 load voltage off	Switch on the load voltage
Main cylinder shut-off valve not opened	Check status of shut-off valve 1V2 MSEB (LED on), check hose at connection 21, Check status output 0 = on Connecting cable output 0 and MSEB 1V2

Gun reports no force reached

Cause	Remedy
Incorrect main cylinder configured	Check the drive data in the project
Supply pressure not sufficient	Check the supply pressure
Gun incorrectly calibrated	Check force calibration

Main cylinder does not reach the expected position

Cause	Remedy
Incorrect main cylinder configured	Check the drive data in the project
Check software limit switch for maximum opening dimension parameter	Check axis calibration
Shut-off valve in main cylinder not opened	Check connection 21 to the main cylinder Check gun ready Check status of output 0 and LED at the shut-off valve 1V2

“Equalizer is back” signal missing

Cause	Remedy
Supply pressure less than 4 bar	Check the supply pressure
Limit switch defective	Check parameterisation and function of limit switch “Equalizer is back”

11 Safety-related specifications in accordance with ISO 13849

11.1 Switching statuses of valves

No supply pressure and load voltage US2 off

- 1V1 at rest (valve in middle position ²⁾ and supply voltage off)
- 1V2 at rest (LED 1V2 off)
- 1V3 at rest (connection 2 blocked at the main cylinder 1A1)
- 1V4 at rest (connection 4 blocked at the main cylinder 1A1)
- 1V5 at rest
- Main cylinder 1A1 pressure free
- Equalizer cylinder 2A1 pressure free

No supply pressure and load voltage US2 on

- 1V1 active regulation of main cylinder 2A1
- 1V2 switched (LED 1V2 on)
- 1V3 at rest (connection 2 blocked at the main cylinder 1A1)
- 1V4 at rest (connection 4 blocked at the main cylinder 1A1)
- 1V5 at rest
- Main cylinder 1A1 pressure free
- Equalizer cylinder 2A1 pressure free

Supply pressure at connection 1 and load voltage US2 off

- 1V1 at rest (valve in middle position ²⁾ and supply voltage off)
- 1V2 at rest (LED 1V2 off)
- 1V3 at rest (connection 2 blocked at the main cylinder 1A1)
- 1V4 at rest (connection 4 blocked at the main cylinder 1A1)
- 1V5 switched
- Main cylinder 1A1 air shut off ¹⁾
- 2V1 active, regulation equalizing cylinder 2A1

¹⁾ The main cylinder 1A1 can drift slowly due to leakage.

²⁾ Middle position through magnet centring

Supply pressure at connection 1 and load voltage US2 on

- 1V1 active, regulation of main cylinder 2A1
- 1V2 switched (LED 1V2 on)
- 1V3 switched
- 1V4 switched
- 1V5 switched
- 2V1 active, regulation equalizing cylinder 2A1

11.2 Safety-engineering block diagrams

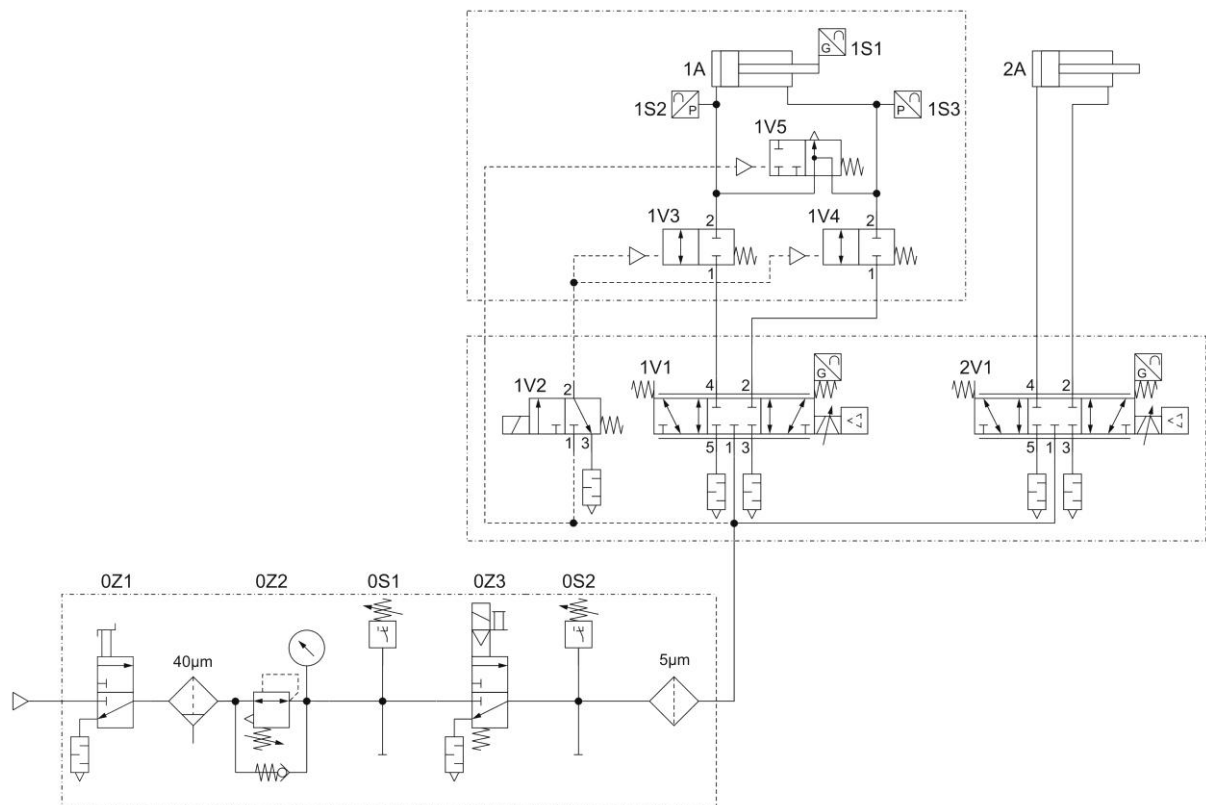


Figure 12: Pneumatic circuit diagram

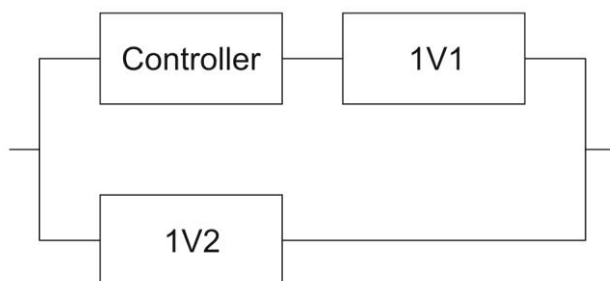


Figure 13: Structure of safety function protection from starting up, load voltage US2 off

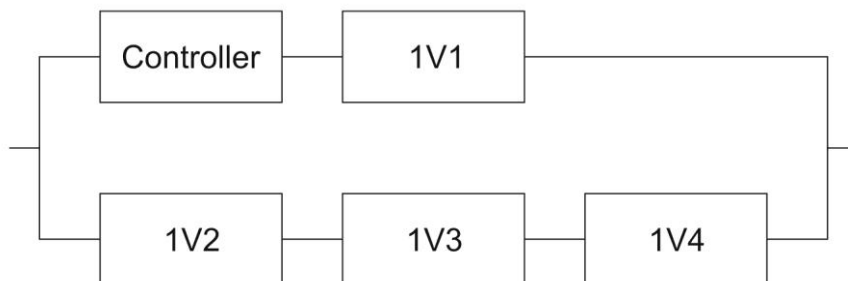


Figure 14: Structure of safety function stop with US2 switch off

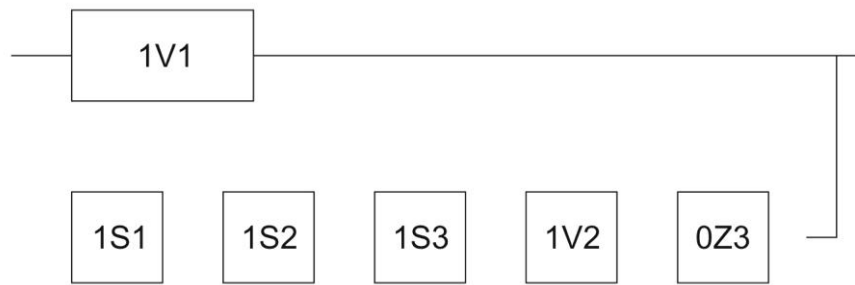


Figure 15: Structure of safety function stop without US2 switch off

11.3 Diagnostic function for error detection of the blocking function in the main cylinder

General

The diagnostic function serves to check the blocking function of the main cylinder 1A1. The function of the valves 1V2, 1V3, 1V4 is checked over a defined process. The diagnostic function is started by the robot. The result of the check is reported to the robot. If the diagnostic function reports an error, it can only be deleted through the error-free sequence of the diagnostic function.

Note:

Diagnostic function is implemented from firmware version V0.60!

Switching off of the supply pressure during execution of the diagnostic function can result in the diagnostic function reporting an error.

Start, diagnostic function

- Robot sets the bits “Diagnostic function” and “Release positioning”.
- Load voltage US2 on.
- Supply pressure at port 1.
- 1V2 at rest (LED 1V2 off).
- 1V3 at rest (connection 2 blocked at the main cylinder 1A1).
- 1V4 at rest (connection 4 blocked at the main cylinder 1A1).
- 1V5 switched.
- 1V1 is activated.

Diagnostic function I/O (blocking function I/O)

- The pressure change and position in the main cylinder 1A1.
- Bit “diagnostic function I/O” = 1.
- 1V2 switched (LED 1V2 on).

Diagnostic function error

- Pressure change or position change in the main cylinder 1A1.
- 1V2 switched (LED 1V2 on).
- Bit “Diagnostic function error” = 1 (bit is saved)
- Bit “Common error” = 1.
- The bit “Diagnostic function error” is reset if the diagnostic function is ended successfully with OK.

A Appendix

A.1 Technical data

General	
Nominal pressure	5 ... 10 bar
Max. operating pressure	12 bar
Permissible overpressure (t= 1 sec, no damage)	16 bar max. 1 sec.
Operating medium at the main air port 1 of the controller	Dried air, unlubricated, in accordance with DIN ISO 8573-1:1995-04 Class 3.4.2 ¹⁾ Pressure dew point at least 10 K below median temperature
Operating voltage – Load voltage supply – Logic supply	PELV circuit in accordance with IEC/DIN EN 60204-1 24 V ±10 % 24 V ±10 %
Operating current (Load and logic supply)	< 4 A, only if no other modules and valves are coupled at the fieldbus node
Max. permissible load current for digital outputs	250 mA
Ambient temperature	0 ... +50 °C
Storage temperature	-10 ... +60 °C
Medium temperature	5 ... +40 °C
Max. permissible relative humidity, non-condensing	90 %
Protection class (DIN EN 60 52)	IP 65 ²⁾
Electromagnetic compatibility – Emitted interference – Resistance to interference	in accordance with EN 61000-6-4: 2007 ³⁾ in accordance with EN 61000-6-2:2005
Shock resistance (IEC68-2-27)	Severity level 1
Vibration resistance (IEC68-2-6)	
¹⁾ A 5 µm filter (OZ1) must be installed in front of the main compressed air connection of the servobox ²⁾ All connections must be sealed with appropriate protective caps. The unused ProfiNet connections must be sealed with appropriate plugs. ³⁾ The component is intended for use in an industrial environment	

Table.3: Technical data

A.2 **Fieldbus protocol**

Bit	Word	Robot outputs	Robot inputs
0	0	Error reset	Ready
1	0	Energy-saving mode On	Energy-saving mode On
2	0	Controller enable off	Controller enable is off
3	0	Without force	With force
4	0	With TCP	Handshake bit TCP OK
5	0	Start learning teach position	
6	0	Teach position is accepted	Teach position valid
7	0	Robot manual	Manual control
8	0	Robot automatic	Automatic control
9	0	Homing	Welding gun is referenced
10	0	Positioning release	Positioning active
11	0	Setpoint position valid	Actual position valid
12	0	Open gun	Gun opened
13	0	Close gun	Force reached
14	0	Gun pushing open	Equalizer is back
15	0	Gun pushing closed	
16	1	Setpoint value E-force W1x100 N	Actual value E-force W1*100 N
17	1	Setpoint value E-force W2x100 N	Actual value E-force W2*100 N
18	1	Setpoint value E-force W4x100 N	Actual value E-force W4*100 N
19	1	Setpoint value E-force W8x100 N	Actual value E-force W8*100 N
20	1	Setpoint value E-force W16x100 N	Actual value E-force W16*100 N
21	1	Setpoint value E-force W32x100 N	Actual value E-force W32*100 N
22	1	Setpoint value E-force W64x100 N	Actual value E-force W64*100 N
23	1	Setpoint value E-force W128x100 N	Actual value E-force W128*100 N
24	1	Equalizer pressure W1	Reset value TCP 1x0.1 mm
25	1	Equalizer pressure W2	Reset value TCP 2x0.1 mm
26	1	Equalizer pressure W4	Reset value TCP 4x0.1 mm
27	1	Equalizer pressure W8	Reset value TCP 8x0.1 mm
28	1	Equalizer pressure W16	Reset value TCP 16x0.1 mm
29	1	Equalizer pressure W32	Reset value TCP 32x0.1 mm
30	1	Equalizer pressure W64	Reset value TCP 64x0.1 mm
31	1	Equalizer pressure W128	Reset value TCP 128x0.1 mm
32	2	Setpoint position W 1x0.1 mm	Actual position W 1x0.1 mm
33	2	Setpoint position W 2x0.1 mm	Actual position W 2x0.1 mm
34	2	Setpoint position W 4x0.1 mm	Actual position W 4x0.1 mm
35	2	Setpoint position W 8x0.1 mm	Actual position W 8x0.1 mm
36	2	Setpoint position W 16x0.1 mm	Actual position W 16x0.1 mm
37	2	Setpoint position W 32x0.1 mm	Actual position W 32x0.1 mm
38	2	Setpoint position W 64x0.1 mm	Actual position W 64x0.1 mm
39	2	Setpoint position W 128x0.1 mm	Actual position W 128x0.1 mm
40	2	Setpoint position W 256x0.1 mm	Actual position W 256x0.1 mm
41	2	Setpoint position W 512x0.1 mm	Actual position W 512x0.1 mm
42	2	Setpoint position W 1024x0.1 mm	Actual position W 1024x0.1 mm

Bit	Word	Robot outputs	Robot inputs
43	2	Setpoint position W 2048x0.1 mm	Actual position W 2048x0.1 mm
44	2	Setpoint position W 4096x0.1 mm	Actual position W 4096x0.1 mm
45	2	Setpoint position W 8192x0.1 mm	Actual position W 8192x0.1 mm
46	2	Setpoint position W 16384x0.1 mm	Actual position W 16384x0.1 mm
47	2	Sign (0=pos. / 1=neg)	Sign (0=pos. / 1=neg)
48	3	Life bit robot	Life bit gun controller
49	3	Compensation robot	
50	3	Caps changed	Cap change OK
51	3	Measure before milling	
52	3	Measure after milling	End measuring
53	3	Start milling completed	
54	3	Normal milling performed	Milling OK
55	3	Function test without component	
56	3	Gun code 1	Invalid gun code
57	3	Gun code 2	
58	3	Gun code 4	
59	3	Gun code reserve	Diagnostic function error
60	3	Diagnostic function	Diagnostic function OK
61	3		Device code W1
62	3		Device code W2
63	3		Device code W4
64	4	Program W1	Preliminary warning gun
65	4	Program W2	Warning, lubrication required
66	4	Program W4	Force build-up too long
67	4	Program W8	Program error / incorrect force specification
68	4	Program W16	Target position invalid
69	4	Program W32	Component failure
70	4	Program W64	Gun moves hard
71	4	Program W128	Milling not OK
72	4	Vehicle type W1	Error cap wear
73	4	Vehicle type W2	Error caps or geometry
74	4	Vehicle type W4	Following error
75	4	Vehicle type W8	Error equalization system
76	4	Vehicle type W16	Error control/regulator
77	4	Vehicle type W32	Error temperature transformer
78	4	Vehicle type W64	Error compressed air supply
79	4	Vehicle type W128	Common malfunction

Table 4: Fieldbus protocol VW Group interface